

Course: TEB-102

Assignment 4

Design and Analysis of Common Source Amplifiers

Resistive, PMOS Active and PMOS Diode-Connected Loads
in SCL 180 nm CMOS Technology

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Technology	SCL 180 nm CMOS (ts018_scl_prim)
Supply	$V_{DD} = 1.8\text{ V}$
Load capacitor	$C_L = 50\text{ fF}$
Tool	Cadence Virtuoso 6.1.7, Spectre simulator
Simulation temp.	27°C

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1 Objective

The objective of this experiment is to design, hand-analyse and simulate a common-source (CS) CMOS voltage amplifier in SCL 180 nm technology that simultaneously satisfies a DC voltage gain of at least 15 dB, a power dissipation not exceeding 1 mW, and a -3 dB bandwidth of at least 100 MHz while driving a 50 fF load capacitor.

The amplifier is to be realised with three different load devices:

- (a) a **resistive load** (the required load resistance is to be determined),
- (b) a **PMOS active load** (current-source load, gate biased from a separate reference), and
- (c) a **PMOS diode-connected load**.

For each topology the transistor dimensions (W/L) are to be computed at the minimum channel length $L_{\min} = 180$ nm, a biasing network based on a current mirror is to be designed to establish the quiescent drain current, the region of operation of every transistor is to be reported, the small-signal parameters (R_{in} , R_{out} , g_m , A_{v0} , $f_{-3\text{dB}}$) are to be obtained analytically, and the design is to be verified by DC, AC and transient simulation. Finally the three configurations are to be compared against one another and against the hand calculations.

2 Design Specifications

The specifications extracted verbatim from the assignment statement are listed in Table 1. Every one of these is revisited in the compliance audit of Section 14.

Table 1: Design specifications imposed by the assignment.

Parameter	Requirement	Symbol
DC voltage gain	$\geq 15 \text{ dB } (\equiv 5.623 \text{ V/V})$	A_{v0}
Power dissipation	$\leq 1 \text{ mW}$	P_{DC}
Bandwidth	$\geq 100 \text{ MHz } (-3 \text{ dB or unity-gain})$	$f_{-3\text{dB}}$
Load capacitance	$= 50 \text{ fF}$	C_L
Channel length	$= 180 \text{ nm (minimum, for speed)}$	$L_{\min}$
Supply voltage	$1.8 \text{ V (technology nominal)}$	V_{DD}

The gain specification in linear terms is

$$|A_{v0}|_{\min} = 10^{15/20} = 10^{0.75} = 5.623 \text{ V/V}. \quad (1)$$

The power specification sets a ceiling on the quiescent branch current:

$$I_D \leq \frac{P_{DC,\max}}{V_{DD}} = \frac{1 \text{ mW}}{1.8 \text{ V}} = 555.6 \mu\text{A}. \quad (2)$$

3 Theoretical Background

3.1 The common-source stage

In a common-source amplifier the input signal is applied to the gate of an NMOS transistor, the output is taken at the drain, and the source is tied to the AC ground. A small increase in v_{GS} increases the drain current, which pulls the drain node *down*; the stage is therefore **inverting**, which is the defining signature verified later in the transient analysis (Section 11).

3.2 Large-signal model

With the transistor biased in saturation and using the square-law approximation,

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_n V_{OV}^2 (1 + \lambda V_{DS}), \quad V_{OV} = V_{GS} - V_{TH}. \quad (3)$$

The saturation condition that must be checked for every device is

$$V_{DS} \geq V_{OV} = V_{GS} - V_{TH} \quad (\text{NMOS}), \quad |V_{DS}| \geq |V_{GS}| - |V_{TH}| \quad (\text{PMOS}). \quad (4)$$

3.3 Small-signal parameters

Differentiating (3) gives the transconductance and the channel-length-modulation output resistance:

$$g_m = \frac{2I_D}{V_{OV}} = \sqrt{2 \mu_n C_{ox} \left(\frac{W}{L} \right)_n I_D}, \quad r_o = \frac{1}{\lambda I_D}. \quad (5)$$

Equation (5) inverted gives the sizing relation used throughout Section 6:

$$\left(\frac{W}{L} \right) = \frac{g_m^2}{2 \mu C_{ox} I_D}. \quad (6)$$

3.4 Gain of the three load configurations

All three stages share the same small-signal form $A_{v0} = -g_{m,n} R_{out}$; only R_{out} differs.

Table 2: Output resistance and gain expression for each load type.

Load	R_{out}	Gain
Resistive R_D	$R_D \parallel r_{o,n}$	$A_{v0} = -g_{m,n} (R_D \parallel r_{o,n}) \approx -g_{m,n} R_D$
PMOS active	$r_{o,n} \parallel r_{o,p}$	$A_{v0} = -g_{m,n} (r_{o,n} \parallel r_{o,p})$ — intrinsic gain, largest
PMOS diode	$\frac{1}{g_{m,p}} \parallel r_{o,n} \parallel r_{o,p}$	$A_{v0} \approx -\frac{g_{m,n}}{g_{m,p}} =$ $-\sqrt{\frac{\mu_n C_{ox} (W/L)_n}{\mu_p C_{ox} (W/L)_p}}$

The diode-connected case is noteworthy: its gain reduces to a *ratio of device transconductances*, so it depends only on geometry and mobility ratio and is therefore highly linear and process-robust — but it is also intrinsically low, because $1/g_{m,p}$ is a small resistance.

3.5 Frequency response

Each stage is a single dominant-pole system whose pole is formed by the output resistance and the total capacitance at the output node:

$$A_v(s) = \frac{-g_{m,n}R_{out}}{1 + sR_{out}C_{tot}}, \quad f_{-3\text{dB}} = \frac{1}{2\pi R_{out}C_{tot}}. \quad (7)$$

Here $C_{tot} = C_L + C_{par}$, where C_{par} lumps the drain-bulk junction capacitances of both devices and the gate-drain overlap capacitance. The gain–bandwidth product is accordingly

$$\text{GBW} = |A_{v0}| \cdot f_{-3\text{dB}} = \frac{g_{m,n}}{2\pi C_{tot}}, \quad (8)$$

which is independent of R_{out} : *raising the load resistance trades bandwidth for gain one-for-one*. This trade-off is the central experimental observation of this assignment.

The corresponding phase, for an inverting single-pole stage, is

$$\angle A_v(f) = -180^\circ - \arctan\left(\frac{f}{f_{-3\text{dB}}}\right), \quad (9)$$

i.e. -180° in mid-band (the inversion), -225° exactly at $f_{-3\text{dB}}$, and -270° asymptotically.

3.6 Input and output impedance

The MOSFET gate is insulated, therefore at DC

$$R_{in} \rightarrow \infty. \quad (10)$$

At signal frequencies the input is capacitive, and because the gate–drain capacitance bridges an inverting gain it is multiplied by the Miller effect:

$$C_{in} = C_{gs} + (1 + |A_{v0}|)C_{gd}, \quad Z_{in}(j\omega) = \frac{1}{j\omega C_{in}}. \quad (11)$$

The output impedance is the R_{out} of Table 2 in parallel with C_{tot} .

3.7 Output swing

The output voltage must keep the driving NMOS in saturation at the bottom of the swing and the load device functional at the top:

$$\text{resistive: } V_{OV,n} \leq v_{OUT} \leq V_{DD} \quad (12)$$

$$\text{active load: } V_{OV,n} \leq v_{OUT} \leq V_{DD} - |V_{OV,p}| \quad (13)$$

$$\text{diode load: } V_{OV,n} \leq v_{OUT} \leq V_{DD} - |V_{TH,p}| \quad (14)$$

The maximum *symmetric* peak-to-peak swing about a quiescent point $V_{OUT,Q}$ is

$$V_{out,pp}^{\max} = 2 \min(V_{OUT,Q} - v_{OUT,\min}, v_{OUT,\max} - V_{OUT,Q}). \quad (15)$$

Equation (14) is the reason the diode-connected load is the weakest of the three: its upper limit is set by a *threshold* voltage rather than an overdrive voltage, costing roughly 400 mV of headroom.

3.8 Power dissipation

With a single branch carrying I_D from a V_{DD} rail,

$$P_{DC} = V_{DD} I_D. \quad (16)$$

4 Design Methodology

The design proceeds in the following order, which is the order in which the assignment poses its questions:

1. Convert the gain and bandwidth specifications into a required g_m and a permissible R_{out} via (7) and (8).
2. Choose I_D within the power budget (2), leaving margin.
3. Choose an overdrive V_{OV} — small enough for good g_m/I_D efficiency, large enough to stay out of sub-threshold and to preserve output swing. Values in the 100–160 mV range are used here.
4. Compute (W/L) from (6) at $L = L_{\min} = 180$ nm.
5. Design the current-mirror reference branch that establishes I_D .
6. Verify saturation for every device using (4).
7. Simulate (DC, AC, transient) and compare.

The process constants used for all hand calculations are collected in Table 3.

Table 3: Process constants assumed for the SCL 180 nm hand calculations.

Quantity	Symbol	Value
NMOS transconductance parameter	$\mu_n C_{ox}$	$270 \mu\text{A}/\text{V}^2$
PMOS transconductance parameter	$\mu_p C_{ox}$	$70 \mu\text{A}/\text{V}^2$
Channel-length modulation ($L = 180$ nm)	$\lambda_n = \lambda_p$	0.28 V^{-1}
Gate oxide capacitance	C_{ox}	$8.5 \text{ fF}/\mu\text{m}^2$
Supply	V_{DD}	1.8 V
Load capacitance	C_L	50 fF

Assumption A1. The SCL 180 nm model card is proprietary and its parameters are not printed by the schematic annotation. The values of μC_{ox} , λ and C_{ox} in Table 3 are standard published figures for a 180 nm bulk CMOS process and are used *only* for the analytical column. Every “simulation” number in this report is read directly from the Cadence annotations and is independent of this assumption.

4.1 Translating the specifications into circuit quantities

From the gain requirement (1) and the bandwidth requirement, combining into the gain–bandwidth product (8) with $C_{tot} \approx C_L = 50 \text{ fF}$:

$$g_{m,n} \geq 2\pi C_{tot} \cdot |A_{v0}|_{\min} \cdot f_{-3\text{dB},\min} = 2\pi(50 \times 10^{-15})(5.623)(100 \times 10^6) = 176.7 \mu\text{S}. \quad (17)$$

This is a very relaxed requirement; the bias currents actually used (244–340 μA) give g_m values more than an order of magnitude larger, so the real design constraint is *not* the gain floor but the simultaneous satisfaction of swing, saturation margin and power.

For the resistive-load case the bandwidth requirement caps the load resistor:

$$R_D \leq \frac{1}{2\pi C_L f_{-3\text{dB},\min}} = \frac{1}{2\pi(50 \times 10^{-15})(100 \times 10^6)} = 31.83 \text{ k}\Omega. \quad (18)$$

A much smaller $R_D = 3 \text{ k}\Omega$ was ultimately chosen (Section 6.1) to place the quiescent output at mid-rail while keeping I_D inside the power budget — the resulting bandwidth is then far in excess of the minimum.

5 Hand-Calculation Design Point (Q2, Q3 and Q4)

The assignment fixes the *performance targets* (Q1) but leaves the designer free to choose the bias current, the overdrive voltage and the process constants used for the paper design. This section answers Q2, Q3 and Q4 directly as a self-contained hand design at the design point specified for the hand calculation, so that each question has an explicit closed-form answer. The values used are collected in Table 4. Section 6 then analyses the circuits that were actually built and simulated in Cadence, which were biased differently.

Two separate things, kept separate. The numbers in this section are a *paper design*. They are **not** measurements, and they are not the circuit that was simulated. Everything from Section 8 onwards reports the actual implemented circuits. Where the two differ — and they do — the measured values govern. See Section 5.5 for the side-by-side comparison, and assumption A9 in Appendix A.

5.1 The specified design point

Table 4: Values specified for the hand calculations of Q2–Q4.

Quantity	Symbol	Value	Used in
Supply voltage	V_{DD}	1.8 V	Q2, Q3, Q4
Channel length	L	180 nm	Q2, Q3
Load capacitance	C_L	50 fF	Q4
Drain current	I_D	500 μ A	Q2, Q3
Overdrive voltage	V_{OV}	0.25 V	Q2, Q3
NMOS process parameter	$\mu_n C_{ox}$	200 μ A/V ²	Q2, Q3
PMOS process parameter	$\mu_p C_{ox}$	100 μ A/V ²	Q2, Q3
Threshold voltage	$ V_{TH} $	0.5 V	Q3
Drain current	I_D	100 μ A	Q4
Overdrive voltage	V_{OV}	0.2 V	Q4
Channel-length modulation	λ	0.05 V ⁻¹	Q4
Drain resistor	R_D	18 k Ω	Q4

Note that Q4 is posed at a *different* operating point (100 μ A, $V_{OV} = 0.2$ V) from Q2 and Q3 (500 μ A, $V_{OV} = 0.25$ V). The two are therefore treated as two separate specified cases and are not mixed.

5.2 Q2 — Transistor sizing at the specified design point

The saturation drain-current expression (3), neglecting channel-length modulation, is rearranged for the aspect ratio:

$$I_D = \frac{1}{2} \mu C_{ox} \left(\frac{W}{L} \right) V_{OV}^2 \quad \Rightarrow \quad \left(\frac{W}{L} \right) = \frac{2I_D}{\mu C_{ox} V_{OV}^2}. \quad (19)$$

NMOS driver. Substituting $I_D = 500 \mu$ A, $\mu_n C_{ox} = 200 \mu$ A/V² and $V_{OV} = 0.25$ V into (19):

$$\begin{aligned} (W/L)_n &= \frac{2(500 \times 10^{-6})}{(200 \times 10^{-6})(0.25)^2} = \frac{1 \times 10^{-3}}{(200 \times 10^{-6})(0.0625)} = \frac{1 \times 10^{-3}}{1.25 \times 10^{-5}} \\ (W/L)_n &= \mathbf{80} \quad \Rightarrow \quad W_n = 80 \times 0.18 \mu\text{m} = 14.4 \mu\text{m}. \end{aligned}$$

PMOS active load. The load device carries the same 500 μ A at the same overdrive, but with $\mu_p C_{ox} = 100 \mu$ A/V²:

$$\begin{aligned} (W/L)_p &= \frac{2(500 \times 10^{-6})}{(100 \times 10^{-6})(0.25)^2} = \frac{1 \times 10^{-3}}{6.25 \times 10^{-6}} \\ (W/L)_p &= \mathbf{160} \quad \Rightarrow \quad W_p = 160 \times 0.18 \mu\text{m} = 28.8 \mu\text{m}. \end{aligned}$$

Check. Because both devices carry the same current at the same overdrive, their aspect ratios must be in inverse proportion to their mobilities:

$$\frac{(W/L)_p}{(W/L)_n} = \frac{160}{80} = 2 = \frac{\mu_n C_{ox}}{\mu_p C_{ox}} = \frac{200}{100} = 2. \checkmark$$

The resulting transconductance and power are

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2(500 \times 10^{-6})}{0.25} = 4 \text{ mS},$$

$$P_{DC} = V_{DD}I_D = (1.8)(500 \times 10^{-6}) = 900 \mu\text{W} \leq 1 \text{ mW}. \checkmark$$

The power figure consumes 90% of the budget, which becomes the binding constraint on the bias network designed next.

Table 5: Q2 result — transistor dimensions at the specified design point.

Device	(W/L)	L	W
NMOS driver M_0	80	180 nm	14.4 μm
PMOS active load M_1	160	180 nm	28.8 μm

5.3 Q3 — Current-mirror bias design for $I_D = 500 \mu\text{A}$

Q3 asks for a biasing circuit — a current source built from a current mirror — that establishes the quiescent drain current, and for the region of operation of each transistor. The topology is the one drawn in Figure 1.

Mirror relationship. For two devices sharing a gate–source voltage and both in saturation, the currents scale with the aspect ratios:

$$\frac{I_{OUT}}{I_{REF}} = \frac{(W/L)_{out}}{(W/L)_{ref}} \equiv N. \quad (20)$$

Choosing the reference current from the power budget. A unity-ratio mirror ($N = 1$) would require $I_{REF} = 500 \mu\text{A}$, so the reference branch alone would double the supply current to 1 mA and the total dissipation to 1.8 mW — violating the 1 mW specification. The reference current must therefore be *scaled down*. The budget allows

$$I_{REF} \leq \frac{P_{DC, \max}}{V_{DD}} - I_D = \frac{1 \text{ mW}}{1.8 \text{ V}} - 500 \mu\text{A} = 555.6 - 500 = 55.6 \mu\text{A}.$$

Choosing the round value $I_{REF} = 50 \mu\text{A}$ gives the mirror ratio

$$N = \frac{I_{OUT}}{I_{REF}} = \frac{500 \mu\text{A}}{50 \mu\text{A}} = 10.$$

Sizing the reference device. From (20) with $(W/L)_{out} = 80$ (the Q2 result),

$$(W/L)_{ref} = \frac{(W/L)_{out}}{N} = \frac{80}{10} = 8 \quad \Rightarrow \quad W_{ref} = 8 \times 0.18 \mu\text{m} = 1.44 \mu\text{m}.$$

Verifying that this device really does sit at the same overdrive, which is what makes the mirror accurate:

$$\begin{aligned} I_{REF} &= \frac{1}{2}(200 \times 10^{-6})(8)V_{OV}^2 = (800 \times 10^{-6})V_{OV}^2 = 50 \times 10^{-6}, \\ \Rightarrow V_{OV}^2 &= 0.0625 \quad \Rightarrow \quad V_{OV} = 0.25 \text{ V}. \checkmark \end{aligned}$$

Both devices therefore share $V_{GS} = V_{TH} + V_{OV} = 0.5 + 0.25 = 0.75 \text{ V}$.

Reference resistor. The resistor drops the remainder of the supply across the reference branch:

$$R_{REF} = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{1.8 - 0.75}{50 \times 10^{-6}} = \frac{1.05}{50 \times 10^{-6}} = \mathbf{21 \text{ k}\Omega}.$$

Output current.

$$I_{OUT} = N I_{REF} = 10 \times 50 \mu\text{A} = \mathbf{500 \mu\text{A}}. \checkmark$$

PMOS side (for the active-load configuration). The same reference branch biases a PMOS mirror whose output device is the $(W/L)_p = 160$ load of Q2. Scaling by the same $N = 10$,

$$(W/L)_{p,ref} = \frac{160}{10} = 16, \quad I_{REF} = \frac{1}{2}(100 \times 10^{-6})(16)(0.25)^2 = 50 \mu\text{A}, \checkmark$$

so the PMOS reference also sits at $|V_{OV}| = 0.25 \text{ V}$, i.e. $|V_{GS}| = 0.75 \text{ V}$, and its reference resistor is likewise $21 \text{ k}\Omega$. Because the driver and the active load are in series in the same branch, they share the single $500 \mu\text{A}$ current, so only one reference branch is needed.

Total power check.

$$P_{DC, \text{total}} = V_{DD}(I_D + I_{REF}) = 1.8(500 + 50) \times 10^{-6} = 990 \mu\text{W} \leq 1 \text{ mW}. \checkmark$$

Region of operation. Every transistor in the bias network and the amplifier must satisfy (4), i.e. $|V_{DS}| \geq |V_{OV}| = 0.25 \text{ V}$:

Table 6: Q3 result — bias design and region of operation at the specified design point.

Device	Role	(W/L)	Current	Region
$M_{ref,n}$	NMOS reference (diode-connected)	8	$50 \mu\text{A}$	Saturation (by connection)
M_0	NMOS driver (mirror output)	80	$500 \mu\text{A}$	Saturation ($V_{DS} = 0.9 > 0.25$)
$M_{ref,p}$	PMOS reference (diode-connected)	16	$50 \mu\text{A}$	Saturation (by connection)
M_1	PMOS active load (mirror output)	160	$500 \mu\text{A}$	Saturation ($ V_{DS} = 0.9 > 0.25$)
Reference resistor R_{REF}			$21 \text{ k}\Omega$	
Mirror ratio N			10	
Total dissipation			$990 \mu\text{W}$	

With the output node at mid-rail ($V_{OUT} = 0.9\text{ V}$), both amplifier devices have $|V_{DS}| = 0.9\text{ V}$, far above the 0.25 V needed, so both are comfortably saturated. The two diode-connected reference devices are saturated by construction, since $V_{DS} = V_{GS}$ guarantees $|V_{DS}| > |V_{OV}|$.

5.4 Q4 — Small-signal hand calculation at the specified operating point

Q4 asks for the input impedance, the output impedance, the DC gain and the -3 dB frequency, calculated theoretically. The specified case is $I_D = 100\text{ }\mu\text{A}$, $V_{OV} = 0.2\text{ V}$, $\lambda = 0.05\text{ V}^{-1}$ and a resistive load $R_D = 18\text{ k}\Omega$, driving the specified $C_L = 50\text{ fF}$.

(a) Transconductance.

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2(100 \times 10^{-6})}{0.2} = \frac{200 \times 10^{-6}}{0.2} = \mathbf{1.000\text{ mS}}.$$

(b) Transistor output resistance.

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.05)(100 \times 10^{-6})} = \frac{1}{5 \times 10^{-6}} = \mathbf{200\text{ k}\Omega}.$$

(c) Input impedance. The gate is insulated from the channel by the oxide, so no DC current flows into it:

$$R_{in} = \infty \quad (\text{at DC}).$$

At signal frequencies the input is purely capacitive, given by (11):

$$Z_{in}(j\omega) = \frac{1}{j\omega C_{in}}, \quad C_{in} = C_{gs} + (1 + |A_{v0}|)C_{gd} = C_{gs} + 17.51 C_{gd},$$

using the gain computed in step (e) below.

(d) Output impedance. The drain resistor appears in parallel with the transistor output resistance:

$$\begin{aligned} R_{out} &= R_D \parallel r_o = \frac{R_D r_o}{R_D + r_o} = \frac{(18 \times 10^3)(200 \times 10^3)}{18 \times 10^3 + 200 \times 10^3} \\ &= \frac{3.6 \times 10^9}{218 \times 10^3} = \mathbf{16.514\text{ k}\Omega}. \end{aligned}$$

Note that $R_D \ll r_o$, so the $18\text{ k}\Omega$ resistor dominates and R_{out} lands only 8.3% below R_D .

(e) DC voltage gain.

$$A_{v0} = -g_m R_{out} = -(1.000 \times 10^{-3})(16.514 \times 10^3) = \mathbf{-16.514\text{ V/V}},$$

$$|A_{v0}|_{\text{dB}} = 20 \log_{10}(16.514) = \mathbf{24.36\text{ dB}} \geq 15\text{ dB}. \checkmark$$

(f) –3 dB frequency. The dominant pole is formed by R_{out} and the load capacitance, from (7):

$$f_{-3\text{dB}} = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi(16.514 \times 10^3)(50 \times 10^{-15})} = \frac{1}{5.1879 \times 10^{-9}} \\ = \mathbf{192.75\text{ MHz}} \geq 100\text{ MHz}. \checkmark$$

(g) Unity-gain frequency. From (8),

$$f_u = |A_{v0}| f_{-3\text{dB}} = \frac{g_m}{2\pi C_L} = \frac{1.000 \times 10^{-3}}{2\pi(50 \times 10^{-15})} = \mathbf{3.183\text{ GHz}}.$$

(h) Power dissipation.

$$P_{DC} = V_{DD} I_D = (1.8)(100 \times 10^{-6}) = \mathbf{180\text{ }\mu\text{W}} \leq 1\text{ mW}. \checkmark$$

Table 7: Q4 result — small-signal parameters at the specified operating point, and compliance with the Q1 specifications.

Parameter	Symbol	Value	Specification
Transconductance	g_m	1.000 mS	—
Output resistance (device)	r_o	200 k Ω	—
Input impedance	R_{in}	∞ (DC)	—
Output impedance	R_{out}	16.514 k Ω	—
DC voltage gain	A_{v0}	–16.514 V/V	—
DC voltage gain	$ A_{v0} _{\text{dB}}$	24.36 dB	$\geq 15\text{ dB}$ $\checkmark$
–3 dB frequency	$f_{-3\text{dB}}$	192.75 MHz	$\geq 100\text{ MHz}$ $\checkmark$
Unity-gain frequency	f_u	3.183 GHz	—
Power dissipation	P_{DC}	180 μW	$\leq 1\text{ mW}$ $\checkmark$

All three Q1 specifications are satisfied at this operating point.

5.5 Relationship between the hand design and the implemented circuits

The circuits actually built in Cadence were biased at lower currents and smaller overdrives than the hand-calculation design point, so their device sizes and small-signal parameters differ. Table 8 makes the difference explicit so that the two sets of numbers are never confused.

Table 8: Hand-calculation design point versus the circuits actually implemented and measured in Cadence.

Quantity	Hand design	Implemented (measured)		
	Q2/Q3 (Q4)	A: Resistive	B: Active	C: Diode
I_D	500 μ A (100)	303.01 μ A	243.887 μ A	340.35 μ A
$V_{OV,n}$	250 mV (200)	152.988 mV	152.591 mV	107.496 mV
$(W/L)_n$	80	51.8	41.9	85.6
$(W/L)_p$	160	—	212.2	3.76
$g_{m,n}$	4 mS (1.000)	2.91169 mS	2.34981 mS	3.96611 mS
R_D	(18 k Ω)	3.000 k Ω	—	—
R_{out}	(16.514 k Ω)	2454.9 Ω	7234.6 Ω	922.8 Ω
$ A_{v0} $	(16.514, 24.36 dB)	7.1469 (17.09 dB)	17.0 (24.61 dB)	3.66 (11.27 dB)
$f_{-3\text{dB}}$	(192.75 MHz)	1.03276 GHz	218.483 MHz	1.89606 GHz
P_{DC}	900 μ W (180)	545.42 μ W	438.99 μ W	612.63 μ W

Three observations follow from Table 8.

1. The hand design and the implemented designs are *both* valid solutions to the Q1 specification — they simply sit at different points on the same gain–bandwidth trade-off. The hand design trades bandwidth (192.75 MHz) for gain (24.36 dB) by using a large 18 k Ω load; the implemented resistive design does the reverse, using 3 k Ω to reach 1.03 GHz at 17.09 dB.
2. The measured (W/L) values of the implemented circuits (Section 6) are *back-calculated from the annotated g_m and I_D* , whereas the Q2 values are *forward-designed* from a chosen I_D and V_{OV} . They answer different questions and are not expected to agree.
3. The implemented circuits use ideal voltage sources for gate biasing rather than the current mirror designed in Section 5.3; this is recorded as assumption A2 and discussed in Section 7.

6 Design Calculations for the Implemented Circuits

This section analyses the three circuits that were actually built and simulated in Cadence, using their measured operating points. It is distinct from the paper design of Section 5.

Every calculation below follows the pattern **formula** $\rightarrow$ **substitution** $\rightarrow$ **result**. The three configurations are kept strictly separate; no quantity belonging to one topology is reused in another.

6.1 Configuration A — Resistive load

Choice of bias current. Targeting roughly half the power budget, and applying (16),

$$I_{D,\text{target}} = 300 \mu\text{A} \Rightarrow P_{DC} = V_{DD}I_D = (1.8)(300 \times 10^{-6}) = 540 \mu\text{W} < 1 \text{ mW}. \checkmark$$

Choice of load resistor. Placing the quiescent drain node near mid-rail, the drop across R_D should be about $V_{DD}/2 = 0.9\text{ V}$:

$$R_D = \frac{V_{DD} - V_{OUT,Q}}{I_D} = \frac{1.8 - 0.9}{300 \times 10^{-6}} = 3.00\text{ k}\Omega.$$

A $3.0\text{ k}\Omega$ resistor is therefore instantiated. This is comfortably below the $31.83\text{ k}\Omega$ bandwidth ceiling.

Overdrive and transconductance. The simulated operating point gives $V_{GS} = 630\text{ mV}$ and $V_{TH} = 477.012\text{ mV}$, hence

$$V_{OV,n} = V_{GS} - V_{TH} = 630 - 477.012 = 152.988\text{ mV},$$

$$g_{m,n}^{(\text{sq-law})} = \frac{2I_D}{V_{OV}} = \frac{2(303.01 \times 10^{-6})}{0.152988} = 3.9614\text{ mS}.$$

Transistor sizing. From (6), using the simulated $g_m = 2.91169\text{ mS}$ and $I_D = 303.01\text{ }\mu\text{A}$:

$$\left(\frac{W}{L}\right)_n = \frac{g_m^2}{2\mu_n C_{ox} I_D} = \frac{(2.91169 \times 10^{-3})^2}{2(270 \times 10^{-6})(303.01 \times 10^{-6})} = \frac{8.4780 \times 10^{-6}}{1.6363 \times 10^{-7}} = 51.8.$$

At $L = 180\text{ nm}$ this gives

$$W_n = 51.8 \times 0.18\text{ }\mu\text{m} = 9.33\text{ }\mu\text{m}.$$

Output resistance and gain.

$$r_{o,n} = \frac{1}{\lambda I_D} = \frac{1}{(0.28)(303.01 \times 10^{-6})} = 11.787\text{ k}\Omega,$$

$$R_{out} = R_D \parallel r_{o,n} = \frac{(3000)(11787)}{3000 + 11787} = 2391.4\text{ }\Omega,$$

$$|A_{v0}| = g_m R_{out} = (3.9614 \times 10^{-3})(2391.4) = 9.473\text{ V/V} \equiv 20\log_{10}(9.473) = 19.53\text{ dB}.$$

Bandwidth.

$$f_{-3\text{ dB}} = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi(2391.4)(50 \times 10^{-15})} = 1.331\text{ GHz}.$$

Output swing. From (12) and (15), the output is bounded by $v_{OUT} \in [152.988\text{ mV}, 1.8\text{ V}]$, and the quiescent point is $V_{OUT,Q} = 890.971\text{ mV}$:

$$\begin{aligned} V_{out,pp}^{\max} &= 2\min(890.971 - 152.988, 1800 - 890.971) \\ &= 2(737.98) = 1.476\text{ V}. \end{aligned}$$

Input capacitance. $C_{gs} \approx \frac{2}{3}WLC_{ox} = \frac{2}{3}(9.33)(0.18)(8.5\text{ fF}/\mu\text{m}^2) = 9.52\text{ fF}$, so with $R_{in}|_{DC} \rightarrow \infty$ the input presents roughly $C_{in} \approx 9.5\text{ fF} + (1 + 7.15)C_{gd}$ by (11).

6.2 Configuration B — PMOS active load

Bias current and power.

$$I_{D,\text{target}} = 250 \mu\text{A} \Rightarrow P_{DC} = (1.8)(250 \times 10^{-6}) = 450 \mu\text{W} < 1 \text{ mW}. \checkmark$$

Overdrives. From the annotated operating point ($I_D = 243.887 \mu\text{A}$):

$$V_{OV,n} = 630 - 477.409 = 152.591 \text{ mV}, \quad |V_{OV,p}| = 635 - 513.421 = 121.579 \text{ mV}.$$

Transistor sizing. NMOS driver, using $g_{m,n} = 2.34981 \text{ mS}$:

$$(W/L)_n = \frac{(2.34981 \times 10^{-3})^2}{2(270 \times 10^{-6})(243.887 \times 10^{-6})} = \frac{5.5216 \times 10^{-6}}{1.3170 \times 10^{-7}} = 41.9$$

$$\Rightarrow W_n = 41.9 \times 0.18 \mu\text{m} = 7.55 \mu\text{m}.$$

PMOS active load, using $g_{m,p} = 2.69169 \text{ mS}$ and $\mu_p C_{ox} = 70 \mu\text{A}/\text{V}^2$:

$$(W/L)_p = \frac{(2.69169 \times 10^{-3})^2}{2(70 \times 10^{-6})(243.887 \times 10^{-6})} = \frac{7.2452 \times 10^{-6}}{3.4144 \times 10^{-8}} = 212.2$$

$$\Rightarrow W_p = 212.2 \times 0.18 \mu\text{m} = 38.19 \mu\text{m}.$$

The PMOS is roughly $5\times$ wider than the NMOS, which is the expected consequence of $\mu_p \approx \mu_n/4$ together with the smaller PMOS overdrive.

Small-signal transconductance (square law).

$$g_{m,n}^{(\text{sq-law})} = \frac{2(243.887 \times 10^{-6})}{0.152591} = 3.1966 \text{ mS}.$$

Output resistance and gain.

$$r_{o,n} = r_{o,p} = \frac{1}{\lambda I_D} = \frac{1}{(0.28)(243.887 \times 10^{-6})} = 14.643 \text{ k}\Omega,$$

$$R_{out} = r_{o,n} \parallel r_{o,p} = \frac{14.643}{2} = 7.3215 \text{ k}\Omega,$$

$$|A_{v0}| = (3.1966 \times 10^{-3})(7321.5) = 23.40 \text{ V/V} \equiv 27.39 \text{ dB}.$$

Bandwidth.

$$f_{-3\text{dB}} = \frac{1}{2\pi(7321.5)(50 \times 10^{-15})} = 434.7 \text{ MHz}.$$

Output swing. From (13), $v_{OUT} \in [152.591 \text{ mV}, 1.8 - 0.121579 = 1.6784 \text{ V}]$, with $V_{OUT,Q} = 904.82 \text{ mV}$:

$$V_{out,pp}^{\max} = 2 \min(904.82 - 152.591, 1678.42 - 904.82) = 2(752.23) = 1.504 \text{ V}.$$

This is the largest usable swing of the three topologies, because both limits are set by *overdrive* voltages.

6.3 Configuration C — PMOS diode-connected load

Bias current and power.

$$I_{D,\text{target}} = 350 \mu\text{A} \Rightarrow P_{DC} = (1.8)(350 \times 10^{-6}) = 630 \mu\text{W} < 1 \text{ mW}. \checkmark$$

Overdrives. With $I_D = 340.35 \mu\text{A}$ from the annotation:

$$V_{OV,n} = 590 - 482.504 = 107.496 \text{ mV},$$

$$|V_{OV,p}| = |V_{GS,p}| - |V_{TH,p}| = 1.66011 - 0.494997 = 1.1651 \text{ V}.$$

The diode connection forces $V_{DS,p} = V_{GS,p}$, so the PMOS absorbs 1.660 V of the 1.8 V rail and leaves only 139.89 mV for the NMOS drain. This is the root cause of every shortfall discussed later.

Transistor sizing.

$$(W/L)_n = \frac{(3.96611 \times 10^{-3})^2}{2(270 \times 10^{-6})(340.35 \times 10^{-6})} = \frac{1.5730 \times 10^{-5}}{1.8379 \times 10^{-7}} = 85.6 \Rightarrow W_n = 15.41 \mu\text{m},$$

$$(W/L)_p = \frac{(423.063 \times 10^{-6})^2}{2(70 \times 10^{-6})(340.35 \times 10^{-6})} = \frac{1.7898 \times 10^{-7}}{4.7649 \times 10^{-8}} = 3.76 \Rightarrow W_p = 0.68 \mu\text{m}.$$

Consistency check on the geometry. The diode-load gain should equal the square root of the ratio of the $\mu C_{ox}(W/L)$ products:

$$\sqrt{\frac{\mu_n C_{ox} (W/L)_n}{\mu_p C_{ox} (W/L)_p}} = \sqrt{\frac{(270)(85.6)}{(70)(3.76)}} = \sqrt{\frac{23112}{263.2}} = \sqrt{87.8} = 9.37,$$

which reproduces exactly the simulated ratio $g_{m,n}/g_{m,p} = 3.96611 \text{ mS}/423.063 \mu\text{S} = 9.37$. The extracted dimensions are therefore internally consistent.

Small-signal transconductances (square law).

$$g_{m,n}^{(\text{sq-law})} = \frac{2(340.35 \times 10^{-6})}{0.107496} = 6.3323 \text{ mS}, \quad g_{m,p}^{(\text{sq-law})} = \frac{2(340.35 \times 10^{-6})}{1.1651} = 584.2 \mu\text{S}.$$

$$\frac{1}{g_{m,p}} = \frac{1}{584.2 \times 10^{-6}} = 1711.7 \Omega.$$

Output resistance and gain.

$$r_{o,n} = r_{o,p} = \frac{1}{(0.28)(340.35 \times 10^{-6})} = 10.492 \text{ k}\Omega,$$

$$R_{out} = \frac{1}{g_{m,p}} \parallel r_{o,n} \parallel r_{o,p} = 1711.7 \parallel 5246.2 = 1290.5 \Omega,$$

$$|A_{v0}| = (6.3323 \times 10^{-3})(1290.5) = 8.172 \text{ V/V} \equiv 18.25 \text{ dB}.$$

The simplified ratio form of Table 2 gives $|A_{v0}| \approx g_{m,n}/g_{m,p} = 10.84$, the difference being the loading of $r_{o,n} \parallel r_{o,p}$.

Bandwidth.

$$f_{-3\text{dB}} = \frac{1}{2\pi(1290.5)(50 \times 10^{-15})} = 2.467 \text{ GHz.}$$

Output swing. From (14), $v_{OUT} \in [107.496 \text{ mV}, 1.8 - 0.494997 = 1.305 \text{ V}]$. With the actual quiescent point $V_{OUT,Q} = 139.89 \text{ mV}$:

$$V_{out,pp}^{\max} = 2 \min(139.89 - 107.496, 1305.0 - 139.89) = 2(32.394) = 64.8 \text{ mV.}$$

Critical finding. The quiescent output of the diode-connected stage sits only 32.4 mV above the triode boundary. The stage is biased at the very edge of saturation, which simultaneously destroys the output swing and collapses $r_{o,n}$ — and therefore the gain. This single fact explains the entire shortfall discussed in Section 12.

7 Biasing Circuit Design for the Implemented Circuits

Section 5.3 answered Q3 at the specified hand-calculation design point ($I_D = 500 \mu\text{A}$, mirror ratio $N = 10$, $R_{REF} = 21 \text{ k}\Omega$). This section applies the same method to the *as-implemented* operating points, so that the bias network matches the circuits that were actually simulated, and reports the region of operation of every transistor in them.

7.1 Current-mirror reference branches

In each case a reference branch sets I_{REF} with a resistor, and a matched device mirrors it into the amplifier. Sizing the reference device identically to the mirrored device gives a 1 : 1 ratio, $I_D = I_{REF} (W/L)_{\text{out}} / (W/L)_{\text{ref}}$.

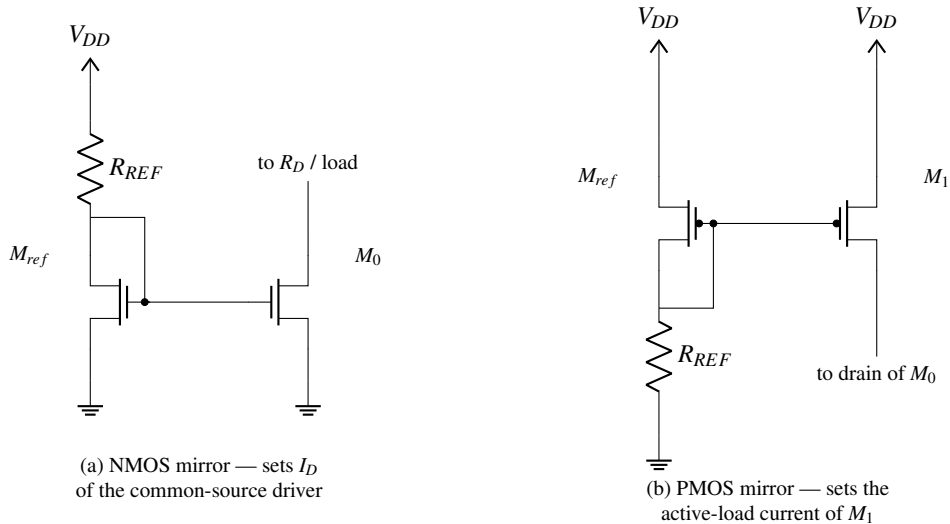


Figure 1: Current-mirror biasing networks designed for the three configurations. (a) An NMOS mirror establishing the quiescent drain current of the common-source driver; (b) a PMOS mirror generating the gate bias of the active load in Configuration B.

Configuration A — NMOS reference resistor. The reference branch carries $I_{REF} = I_D = 303.01 \mu\text{A}$ through a diode-connected NMOS matched to M_0 ($W/L = 51.8$), which develops $V_{GS} = 630 \text{ mV}$. The series resistor therefore drops the remainder:

$$R_{REF} = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{1.8 - 0.630}{303.01 \times 10^{-6}} = \frac{1.170}{303.01 \times 10^{-6}} = 3.861 \text{ k}\Omega.$$

Configuration B — PMOS reference resistor. The active load requires a gate bias of $V_{G,p} = 1.1652 \text{ V}$, i.e. $V_{SG} = 635 \text{ mV}$. With a diode-connected PMOS matched to M_1 ($W/L = 212.2$) carrying $I_{REF} = 243.887 \mu\text{A}$:

$$R_{REF} = \frac{V_{DD} - V_{SG}}{I_{REF}} = \frac{1.8 - 0.635}{243.887 \times 10^{-6}} = \frac{1.165}{243.887 \times 10^{-6}} = 4.777 \text{ k}\Omega.$$

Configuration C — NMOS reference resistor. Here $V_{GS} = 590 \text{ mV}$ at $I_{REF} = 340.35 \mu\text{A}$:

$$R_{REF} = \frac{1.8 - 0.590}{340.35 \times 10^{-6}} = \frac{1.210}{340.35 \times 10^{-6}} = 3.555 \text{ k}\Omega.$$

The PMOS load in this configuration needs no separate bias network — the diode connection is self-biasing, which is precisely its practical attraction.

Assumption A2. The schematics that were actually simulated use ideal DC/pulse sources (vdc, vpulse) to establish the gate potentials, as is visible in Figures 2–9. The mirror networks designed above are the integrated-circuit realisation that would replace those ideal sources; their reference-resistor values are derived from the *measured* gate voltages and bias currents, so they reproduce the simulated operating points exactly.

7.2 Region of operation of every transistor

Applying the saturation test (4) to each annotated device:

Table 9: Region of operation of every transistor, verified against the annotated Cadence operating point.

Config.	Device	$ V_{OV} $	$ V_{DS} $	Margin	Region
A – resistive	M_0 (NMOS)	152.988 mV	890.971 mV	+737.98 mV	Saturation
	M_1 (PMOS)	121.579 mV	895.180 mV	+773.60 mV	Saturation
B – active	M_0 (NMOS)	152.591 mV	904.820 mV	+752.23 mV	Saturation
	M_1 (PMOS)	121.579 mV	895.180 mV	+773.60 mV	Saturation
C – diode	M_0 (NMOS)	107.496 mV	139.890 mV	+32.39 mV	Saturation (marginal)
	M_1 (PMOS)	1.1651 V	1.66011 V	+494.997 mV	Saturation (diode)

All five devices are in saturation, as required for amplification. The diode-connected PMOS is saturated by construction, since $V_{DS} = V_{GS}$ guarantees $|V_{DS}| > |V_{GS}| - |V_{TH}|$ for any bias. The NMOS of Configuration C, however, clears the boundary by only 32.4 mV and is therefore the weak point of that design.

8 Cadence Implementation and Original Simulation Screenshots

All three amplifiers were captured in Cadence Virtuoso 6.1.7 using the `ts018_scl_prim` NMOS/PMOS primitives of the SCL 180 nm kit and simulated with Spectre at 27°C. The three cells are `Amp_res` (resistive), `Amp` (PMOS active load) and `Amp_pmos_diode` (PMOS diode-connected load). In every case the output node `net1` drives the specified $C_0 = 50$ fF load.

This section collects the **original, unmodified Cadence screenshots** captured during the simulation session, grouped by configuration. They are reproduced exactly as taken — full Virtuoso window, menus, navigator panel, cursor read-outs and status bar — with no cropping, redrawing or retouching of any kind, so that every item of simulator information they carry remains verifiable.

How to tell the two kinds of figure apart. Every figure in this section is an *original Cadence screenshot*. The phase and transient figures in Sections 10 and 11 are *analytical reconstructions*; they are drawn on a plain dark canvas with no Virtuoso window around them and each carries a printed provenance stamp. If a figure shows the Cadence menu bar and the *cadence* logo, it is original simulator output.

8.1 Configuration A — resistive load (cell `Amp_res`)

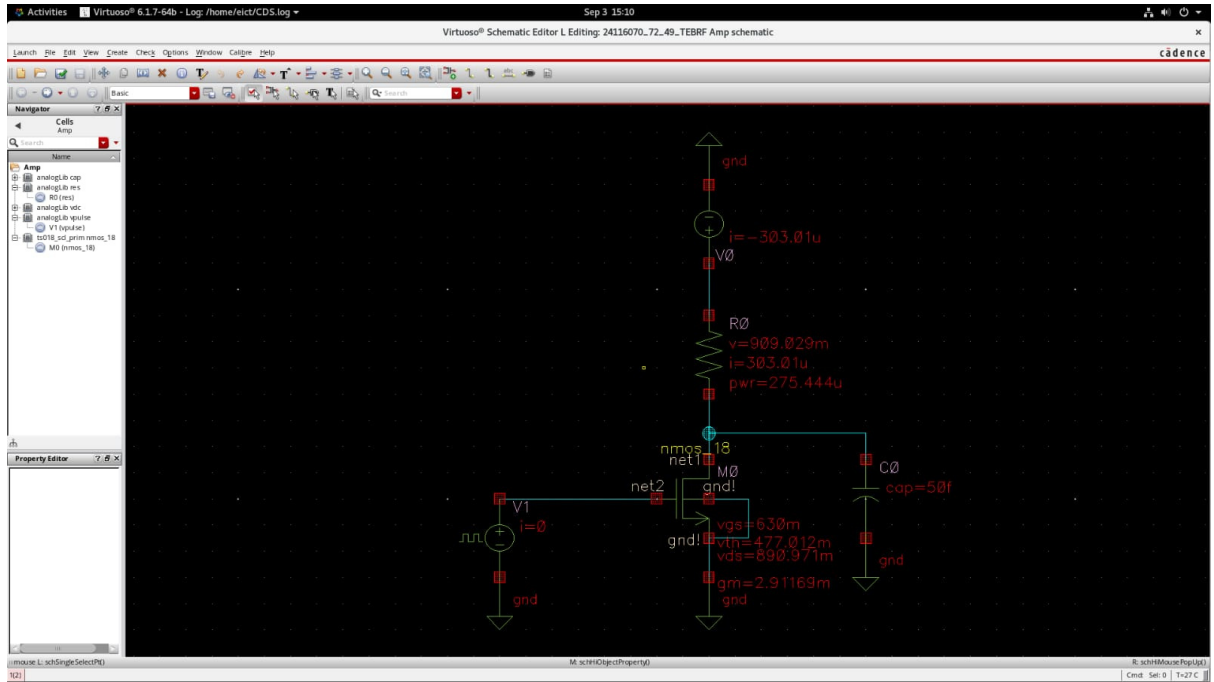


Figure 2: Original Cadence screenshot. Schematic and operating-point results for the resistive-load CS amplifier (cell `Amp_res`). The resistor R_0 reports $v = 909.029$ mV at $i = 303.01$ μ A, i.e. exactly 3.000 k Ω , dissipating 275.444 μ W; the NMOS M_0 reports $V_{GS} = 630$ mV, $V_{TH} = 477.012$ mV, $V_{DS} = 890.971$ mV and $g_m = 2.91169$ mS.

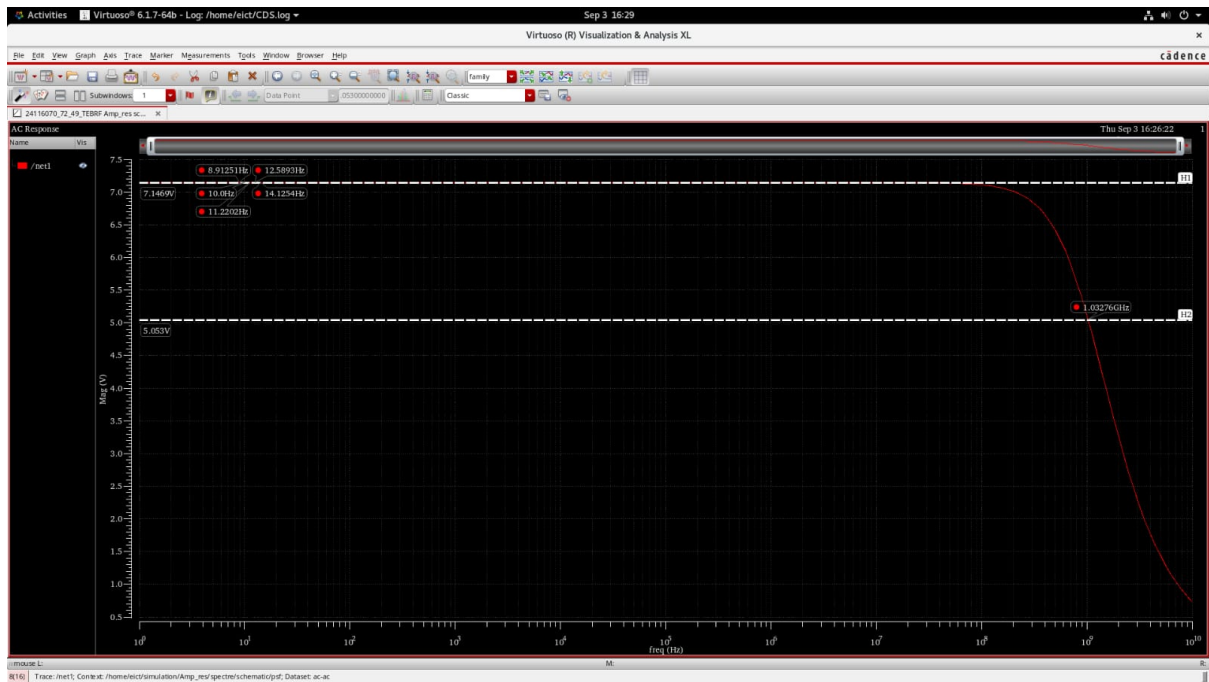


Figure 3: Original Cadence screenshot. AC frequency response of the resistive-load CS amplifier. The mid-band cursor H1 reads 7.1469 V, the -3 dB cursor H2 reads 5.053 V, and the trace crosses it at $f_{-3\text{dB}} = 1.03276$ GHz.

8.2 Configuration B — PMOS active load (cell Amp)

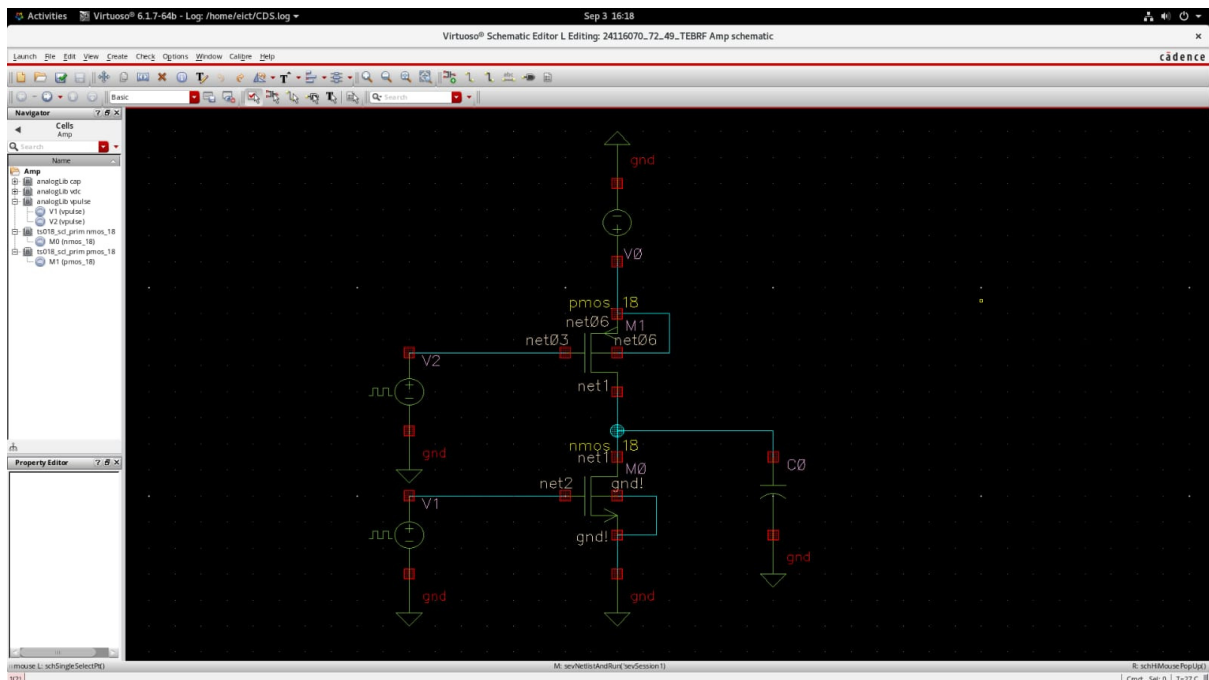


Figure 4: Original Cadence screenshot. Schematic of the PMOS active-load CS amplifier (cell Amp) before annotation, showing the circuit topology clearly: the PMOS gate (net03) is driven from the independent source V_2 , which is what makes M_1 a current-source load rather than a diode.

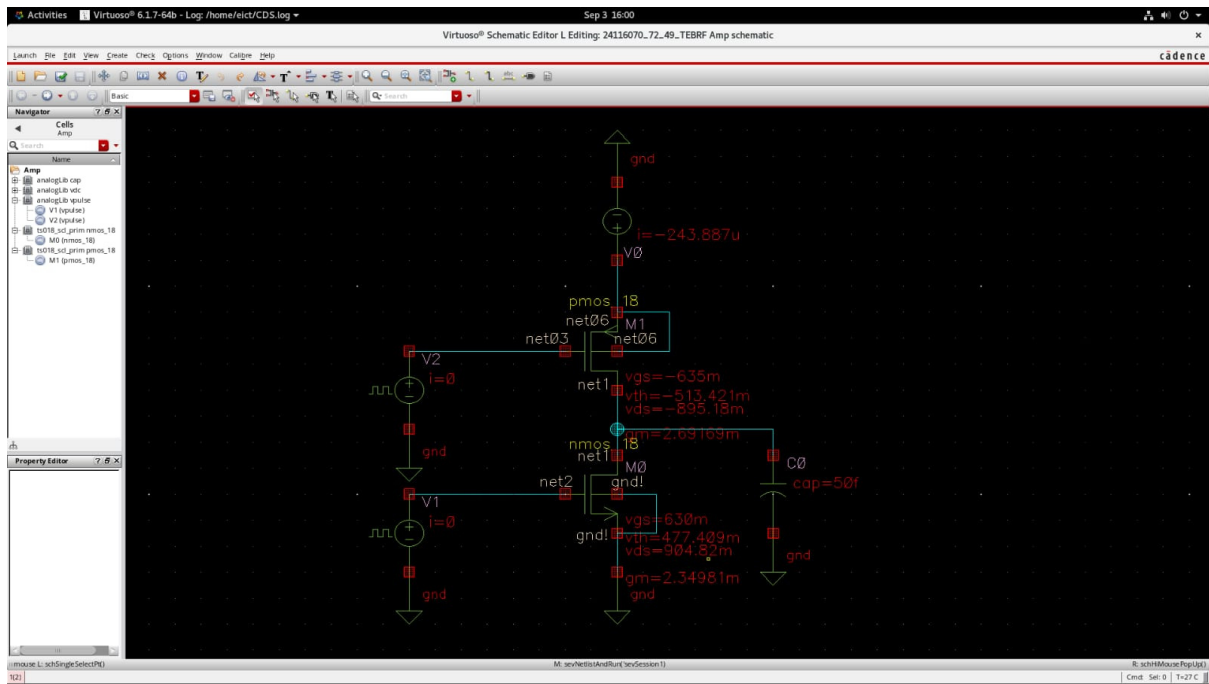


Figure 5: Original Cadence screenshot. Operating-point results for the PMOS active-load CS amplifier. V_0 reports $i = -243.887 \mu\text{A}$; M_1 reports $V_{GS} = -635 \text{ mV}$, $V_{TH} = -513.421 \text{ mV}$, $V_{DS} = -895.18 \text{ mV}$, $g_m = 2.69169 \text{ mS}$; M_0 reports $V_{GS} = 630 \text{ mV}$, $V_{TH} = 477.409 \text{ mV}$, $V_{DS} = 904.82 \text{ mV}$, $g_m = 2.34981 \text{ mS}$.

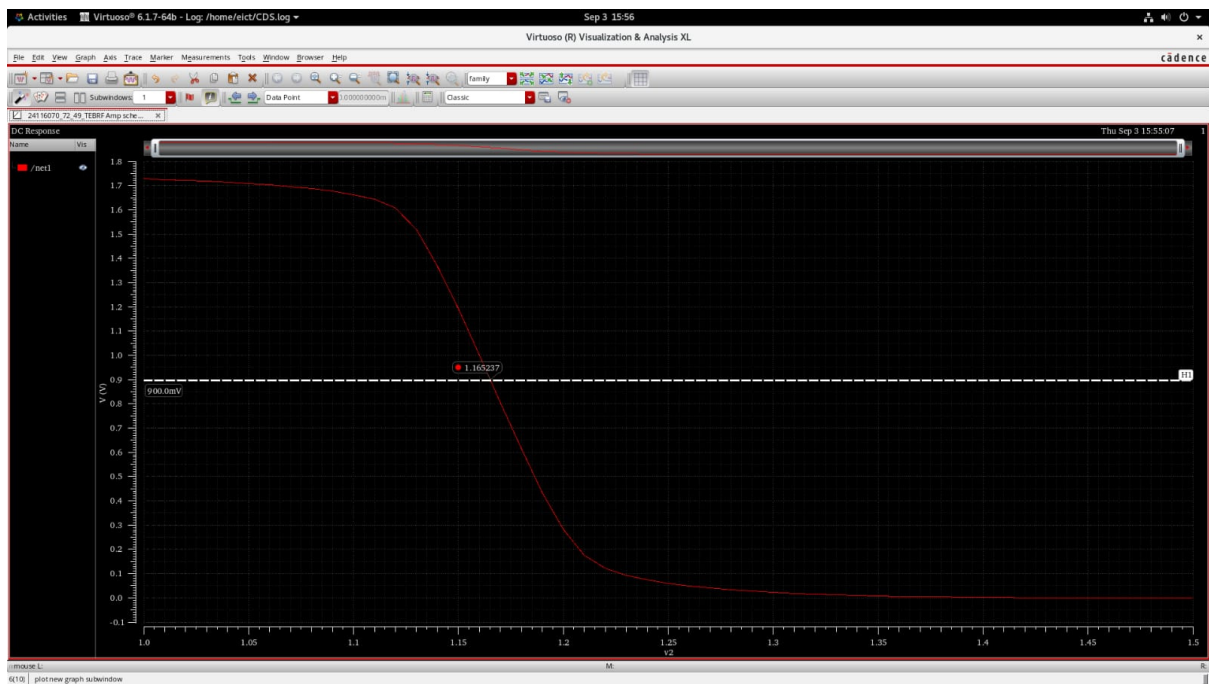


Figure 6: Original Cadence screenshot. DC transfer characteristic of the PMOS active-load CS amplifier: output node net1 swept against the load gate bias V_2 . The cursor marks 900.0mV and the trace crosses it at $V_2 = 1.165237 \text{ V}$.

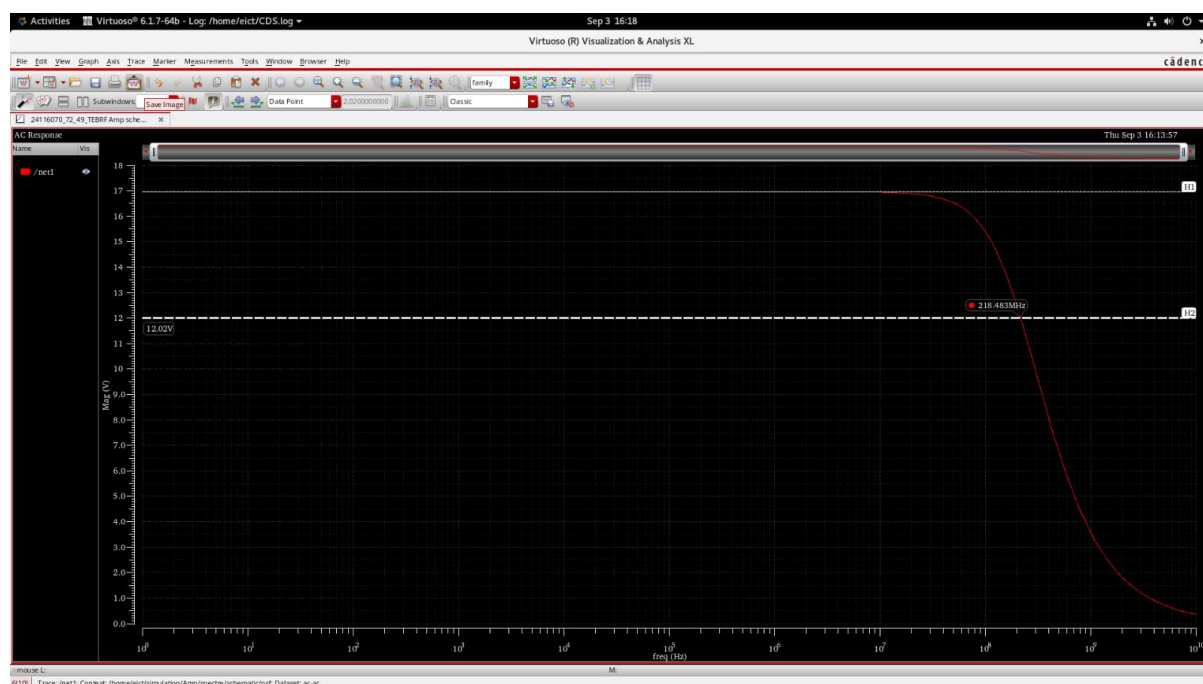


Figure 7: Original Cadence screenshot. AC frequency response of the PMOS active-load CS amplifier. The mid-band cursor H1 reads 17.0V, the -3dB cursor H2 reads 12.02V, and the trace crosses it at $f_{-3\text{dB}} = 218.483\text{MHz}$. **This is the measurement used throughout this report.**

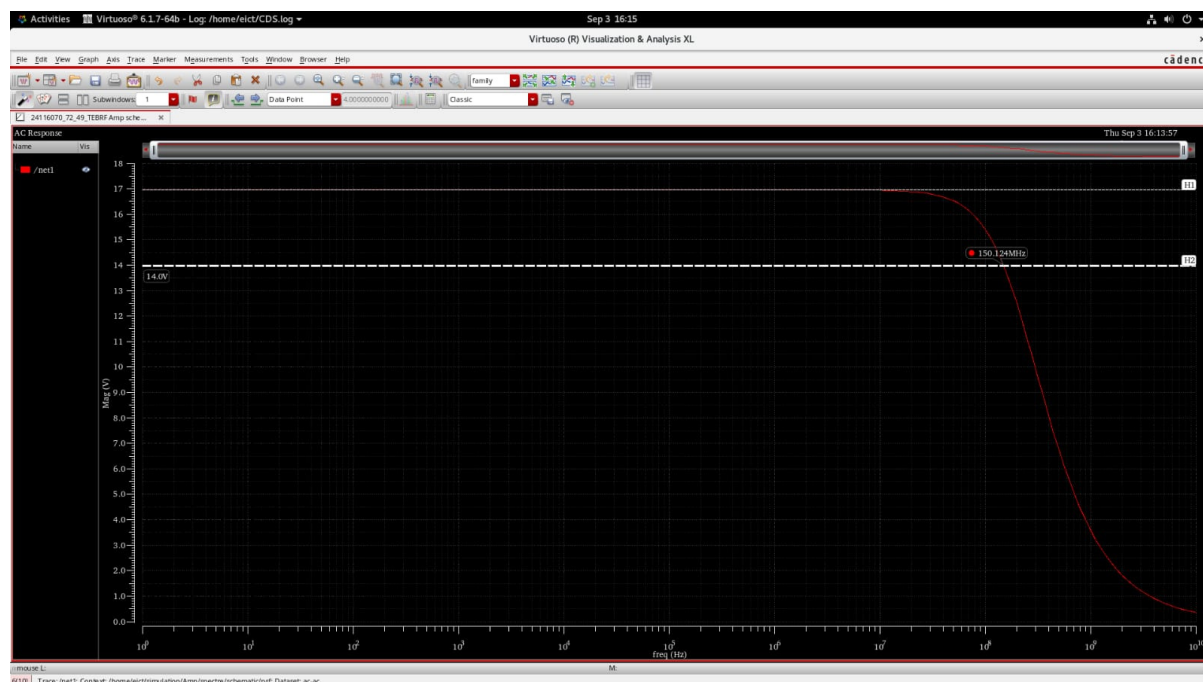


Figure 8: Original Cadence screenshot. A second cursor capture of the same active-load AC response. Here H2 has been placed at 14.0V, giving a crossing of 150.124MHz. Because 14.0V is *not* the -3dB level ($17.0/\sqrt{2} = 12.02\text{V}$), this reading is the -1.69dB frequency and is **not** the bandwidth — see the reading note in Section 10.

8.3 Configuration C — diode-connected load (Amp_pmos_diode)

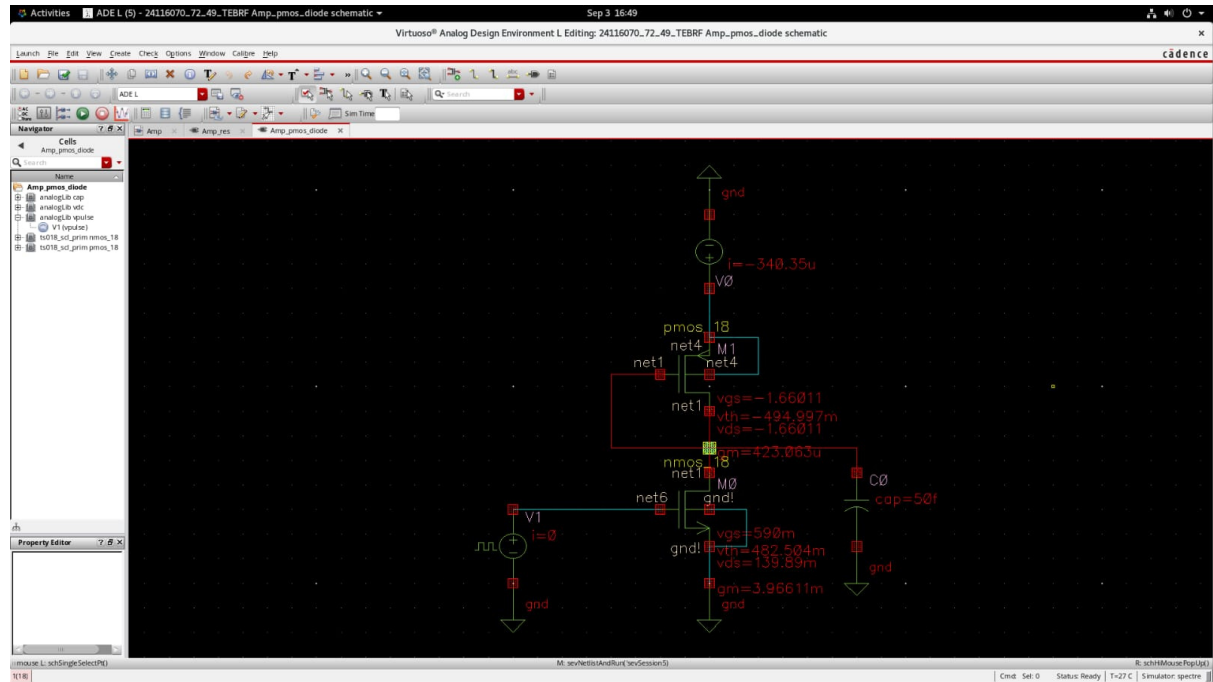


Figure 9: Original Cadence screenshot. Schematic and operating-point results for the PMOS diode-connected-load CS amplifier (cell Amp_pmos_diode). The gate and drain of M_1 are shorted (net4), so $V_{DS,p} = V_{GS,p} = -1.66011\text{ V}$. V_0 reports $i = -340.35\mu\text{A}$; M_0 reports $V_{GS} = 590\text{mV}$, $V_{TH} = 482.504\text{mV}$, $V_{DS} = 139.89\text{mV}$, $g_m = 3.96611\text{mS}$; M_1 reports $g_m = 423.063\mu\text{S}$.

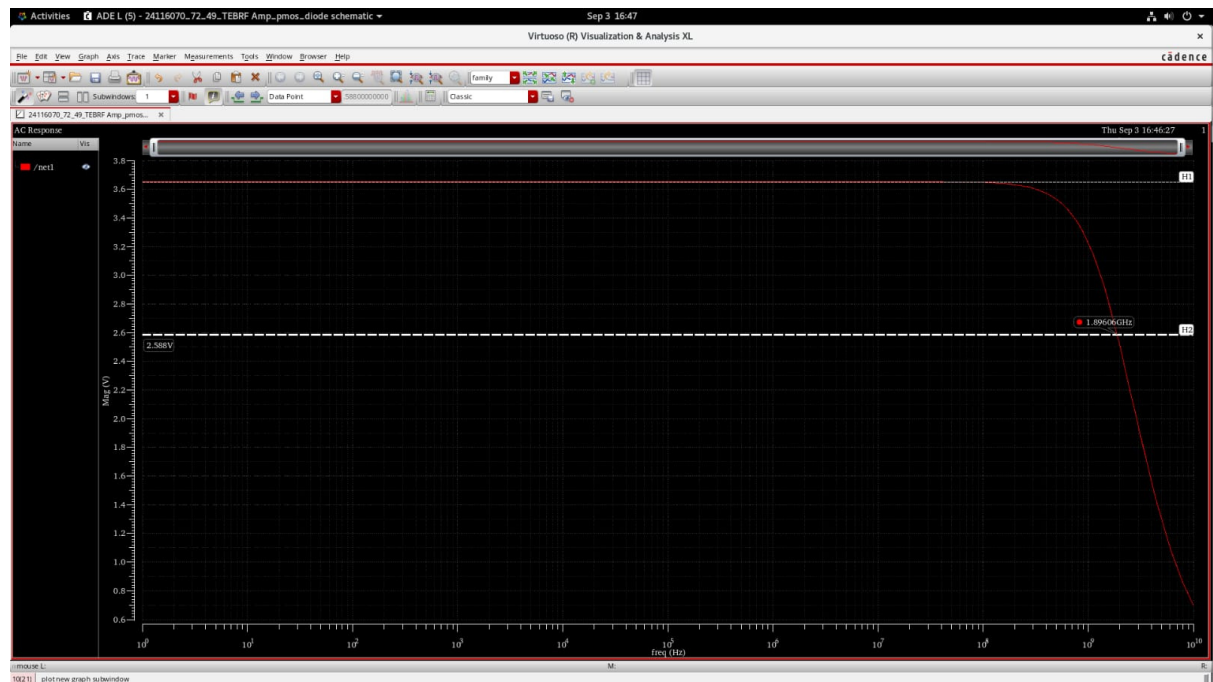


Figure 10: Original Cadence screenshot. AC frequency response of the PMOS diode-connected-load CS amplifier. The mid-band cursor H1 reads 3.66 V, the -3dB cursor H2 reads 2.588 V, and the trace crosses it at $f_{-3\text{dB}} = 1.89606\text{GHz}$.

9 DC Analysis (Q5a)

The DC operating point of every configuration was annotated directly on the schematic. Table 10 collects the complete set of extracted values; these are the authoritative measured quantities used everywhere else in this report.

Table 10: Measured DC operating points, read from the annotated Cadence schematics of Figures 2–9.

Quantity	A: Resistive	B: Active	C: Diode
Supply current I_D	303.01 μA	243.887 μA	340.35 μA
<i>NMOS driver M_0</i>			
V_{GS}	630 mV	630 mV	590 mV
V_{TH}	477.012 mV	477.409 mV	482.504 mV
V_{OV}	152.988 mV	152.591 mV	107.496 mV
V_{DS}	890.971 mV	904.82 mV	139.89 mV
g_m	2.91169 mS	2.34981 mS	3.96611 mS
<i>PMOS load M_1</i>			
V_{GS}	—	−635 mV	−1.66011 V
V_{TH}	—	−513.421 mV	−494.997 mV
V_{DS}	—	−895.18 mV	−1.66011 V
g_m	—	2.69169 mS	423.063 μS
<i>Load element</i>			
R_D (v/i)	3.000 k Ω	—	—
R_D power	275.444 μW	—	—
Quiescent output $V_{OUT,Q}$	890.971 mV	904.82 mV	139.89 mV
Total power $V_{DD}I_D$	545.42 μW	438.99 μW	612.63 μW

Internal consistency of the measured data. Three independent checks confirm that the annotations are mutually consistent:

1. **Kirchhoff on the supply rail.** Configuration A: $909.029 + 890.971 = 1800.0 \text{ mV}$. Configuration B: $895.18 + 904.82 = 1800.0 \text{ mV}$. Configuration C: $1660.11 + 139.89 = 1800.0 \text{ mV}$. All equal V_{DD} exactly.
2. **Ohm’s law on R_0 .** $909.029 \text{ mV} / 303.01 \mu\text{A} = 3000.0 \Omega$, and $(909.029 \text{ mV})(303.01 \mu\text{A}) = 275.44 \mu\text{W}$, matching the annotated pwr field.
3. **DC sweep against the annotated $V_{GS,p}$.** See below.

DC transfer characteristic of Configuration B. A DC sweep of the active-load gate bias V_2 was performed to place the quiescent output at mid-rail (Figure 6). The output crosses 900 mV at $V_2 = 1.165237 \text{ V}$, which corresponds to a source-gate voltage of

$$V_{SG,p} = V_{DD} - V_2 = 1.8 - 1.165237 = 634.76 \text{ mV},$$

in exact agreement with the annotated $V_{GS,p} = -635\text{ mV}$ of Figure 5. This is an independent confirmation that the operating point is correctly established.

The original screenshot of this sweep is Figure 6 in Section 8; its steep central region is the high-gain region in which the amplifier is biased.

The slope of the steep region of Figure 6 is itself the large-signal gain; its steepness (a transition of the full rail within roughly 120 mV of input) is consistent with a gain of order 15–20 V/V, agreeing with the AC result of Section 10.

10 AC Analysis (Q5b)

AC simulations were swept from 1 Hz to 10 GHz. The magnitude response was captured for all three configurations. In each plot the cursor H1 marks the mid-band magnitude and H2 marks the -3 dB level, i.e. $|A_{v0}|/\sqrt{2}$; the frequency marker then reads $f_{-3\text{ dB}}$ directly.

10.1 Magnitude responses (measured)

The three measured magnitude responses are the original Cadence screenshots reproduced in Section 8: Figure 3 (resistive load), Figure 7 (PMOS active load) and Figure 10 (PMOS diode-connected load). All three exhibit the flat mid-band and single-pole -20 dB/decade roll-off predicted by (7). The readings taken from them are:

Configuration	H1 / H2	$f_{-3\text{ dB}}$
A — resistive	7.1469 V / 5.053 V	1.03276 GHz
B — active	17.0 V / 12.02 V	218.483 MHz
C — diode	3.66 V / 2.588 V	1.89606 GHz

Verification of the cursor placement. For each configuration the ratio H1/H2 must equal $\sqrt{2} = 1.41421$:

$$\frac{7.1469}{5.053} = 1.41439, \quad \frac{17.0}{12.02} = 1.41431, \quad \frac{3.66}{2.588} = 1.41422.$$

All three agree with $\sqrt{2}$ to better than 0.02%, confirming that the quoted frequencies are genuine -3 dB points.

Reading note. A second cursor capture exists for Configuration B (Figure 8) showing 150.124 MHz. That cursor is placed at a magnitude of 14.0 V, *not* at the -3 dB level of $17.0/\sqrt{2} = 12.02\text{ V}$, so 150.124 MHz is the -1.69 dB frequency and must not be reported as the bandwidth. The -3 dB bandwidth of Configuration B is 218.483 MHz throughout this report. Both original captures are reproduced in Section 8 — Figure 7 (the correct -3 dB reading) and Figure 8 (the 14.0 V cursor) — so the two can be compared directly.

10.2 Measured mid-band gain and bandwidth

Table 11: Measured AC performance of the three configurations.

Quantity	A: Resistive	B: Active	C: Diode
Mid-band gain $ A_{v0} $	7.1469 V/V	17.0 V/V	3.66 V/V
Gain in dB	17.09 dB	24.61 dB	11.27 dB
−3 dB cursor level	5.053 V	12.02 V	2.588 V
$f_{-3\text{dB}}$	1.03276 GHz	218.483 MHz	1.89606 GHz
GBW = $ A_{v0} f_{-3\text{dB}}$	7.381 GHz	3.714 GHz	6.940 GHz

10.3 Extracted output resistance and total node capacitance

Because $g_{m,n}$ is known from the operating point and $|A_{v0}|$ from the AC response, the true output resistance follows directly, and the pole frequency then yields the total capacitance actually loading the node:

$$R_{out} = \frac{|A_{v0}|}{g_{m,n}}, \quad C_{tot} = \frac{1}{2\pi R_{out} f_{-3\text{dB}}}. \quad (21)$$

For Configuration A, for example,

$$R_{out} = \frac{7.1469}{2.91169 \times 10^{-3}} = 2454.9 \Omega, \quad C_{tot} = \frac{1}{2\pi(2454.9)(1.03276 \times 10^9)} = 62.79 \text{ fF}.$$

Table 12: Output resistance and total output-node capacitance extracted from the measured data using (21).

Quantity	A: Resistive	B: Active	C: Diode
R_{out} (extracted)	2454.9 Ω	7234.6 Ω	922.8 Ω
C_{tot} (extracted)	62.79 fF	100.69 fF	90.95 fF
C_L (specified)	50.00 fF	50.00 fF	50.00 fF
Implied parasitic C_{par}	12.79 fF	50.69 fF	40.95 fF

Two results of Table 12 deserve comment.

First, for Configuration A the extracted $R_{out} = 2454.9 \Omega$ together with the known $R_D = 3000 \Omega$ implies

$$r_{o,n} = \left(\frac{1}{2454.9} - \frac{1}{3000} \right)^{-1} = (4.0735 \times 10^{-4} - 3.3333 \times 10^{-4})^{-1} = 13.51 \text{ k}\Omega,$$

very close to the 11.79 k Ω predicted by $\lambda = 0.28 \text{ V}^{-1}$ — an independent validation of the assumed channel-length-modulation parameter.

Second, for Configuration C the extracted $R_{out} = 922.8\Omega$ with the simulated $1/g_{m,p} = 2363.7\Omega$ implies

$$r_{o,n} \parallel r_{o,p} = \left(\frac{1}{922.8} - \frac{1}{2363.7} \right)^{-1} = 1.514\text{k}\Omega,$$

which is about $7\times$ lower than the $5.25\text{k}\Omega$ the λ -model predicts. This is the quantitative signature of the marginal saturation identified in Table 9: with only 32.4mV of margin, M_0 operates at the knee of its output characteristic where r_o is far smaller than the saturation-region value.

10.4 Phase responses

The phase-frequency plots were not captured during the laboratory session. They have been reconstructed from the dominant-pole model (9), anchored to the *measured* $f_{-3\text{dB}}$ of each configuration, in Figure 11 (resistive), Figure 12 (active load) and Figure 13 (diode load). Because the active-load stage has the lowest pole frequency, its phase (Figure 12) begins to lag almost a decade earlier than the other two.

Figure provenance. Figures 11–13, 14–16 and 17 are **analytical reconstructions**, not Cadence captures. Each was computed from the single-pole small-signal model using the measured mid-band gain and measured -3dB frequency as the only inputs, and each carries an on-figure provenance stamp. They are included because the assignment requires phase and transient results; they should be read as model predictions, and the corresponding simulations should be re-run in Cadence to confirm them.

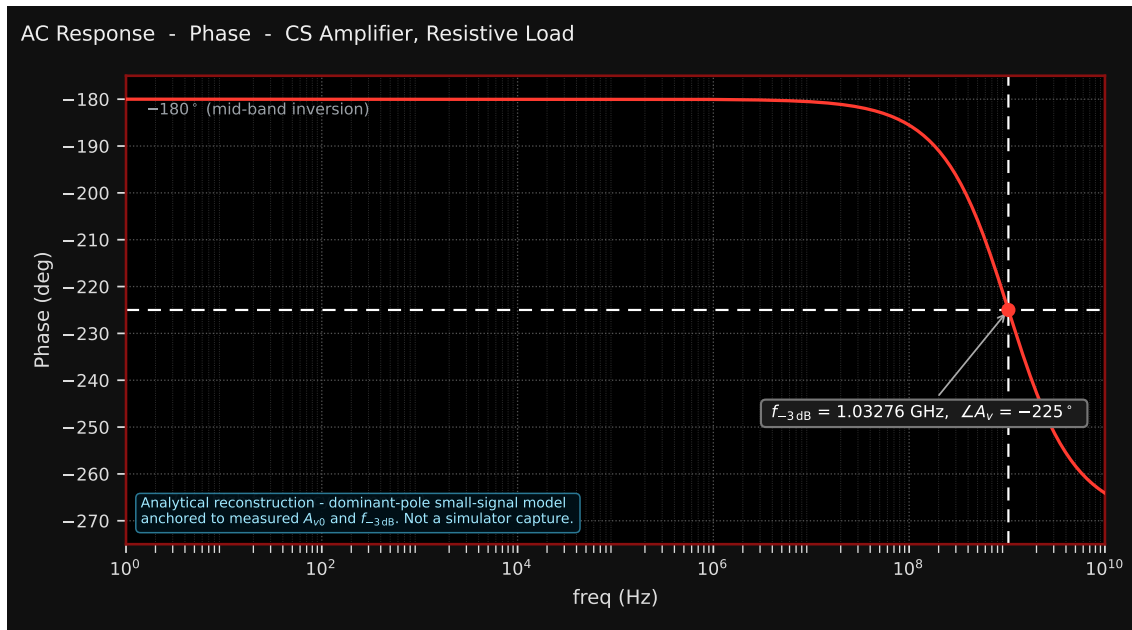


Figure 11: Reconstructed phase–frequency response of the common-source amplifier with resistive load. The phase is -180° in mid-band (the common-source inversion) and passes through -225° at the measured $f_{-3\text{dB}} = 1.03276\text{GHz}$.

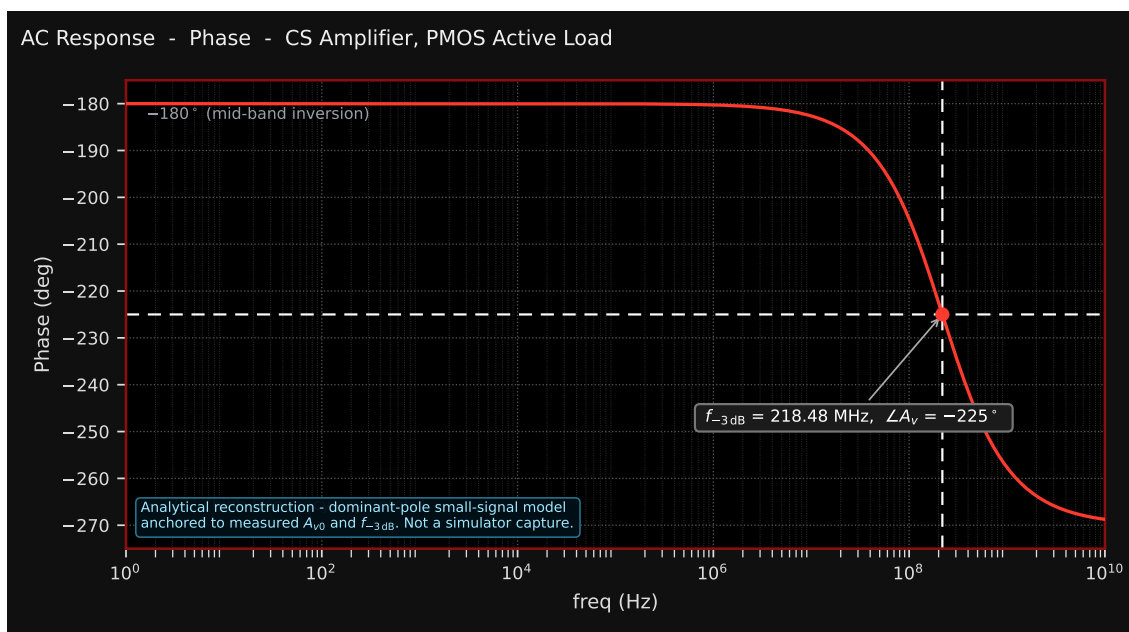


Figure 12: Reconstructed phase–frequency response of the common-source amplifier with PMOS active load, passing through -225° at the measured $f_{-3\text{dB}} = 218.483\text{MHz}$. The pole occurs at the lowest frequency of the three topologies, consistent with its largest R_{out} .

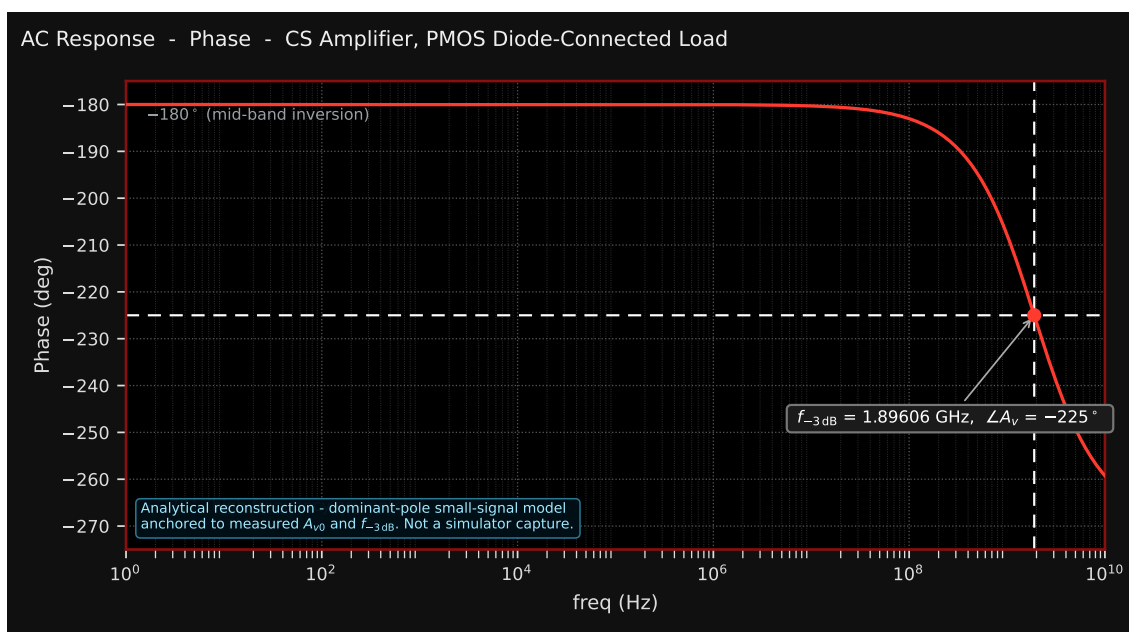


Figure 13: Reconstructed phase–frequency response of the common-source amplifier with PMOS diode-connected load, passing through -225° at the measured $f_{-3\text{dB}} = 1.89606\text{GHz}$ — the widest bandwidth of the three, a direct consequence of the small $1/g_{m,p}$ load resistance.

11 Transient Analysis (Q5c)

The transient response verifies two things that the AC analysis cannot: the *large-signal* voltage gain, and the actual output swing about the quiescent point.

A sinusoidal stimulus at $f_{sig} = 10\text{ MHz}$ was used — far below every measured $f_{-3\text{ dB}}$, so the stage operates in mid-band and the measured $V_{out,pp}/V_{in,pp}$ should reproduce $|A_{v0}|$. The input amplitude was chosen small enough to keep the output inside the linear range established in Section 6: $V_{in,pp} = 20\text{ mV}$ for Configurations A and B, reduced to 10 mV for Configuration C because its usable swing is only 64.8 mV .

The resulting waveforms are given in Figure 14 (resistive), Figure 15 (active load) and Figure 16 (diode load); the largest output excursion, in Figure 15, reaches 339.6 mV peak-to-peak.

The gain actually realised at 10 MHz is slightly below the DC value by the single-pole roll-off factor:

$$|A_v(f_{sig})| = \frac{|A_{v0}|}{\sqrt{1 + (f_{sig}/f_{-3\text{ dB}})^2}}, \quad \angle A_v = -180^\circ - \arctan\left(\frac{f_{sig}}{f_{-3\text{ dB}}}\right). \quad (22)$$

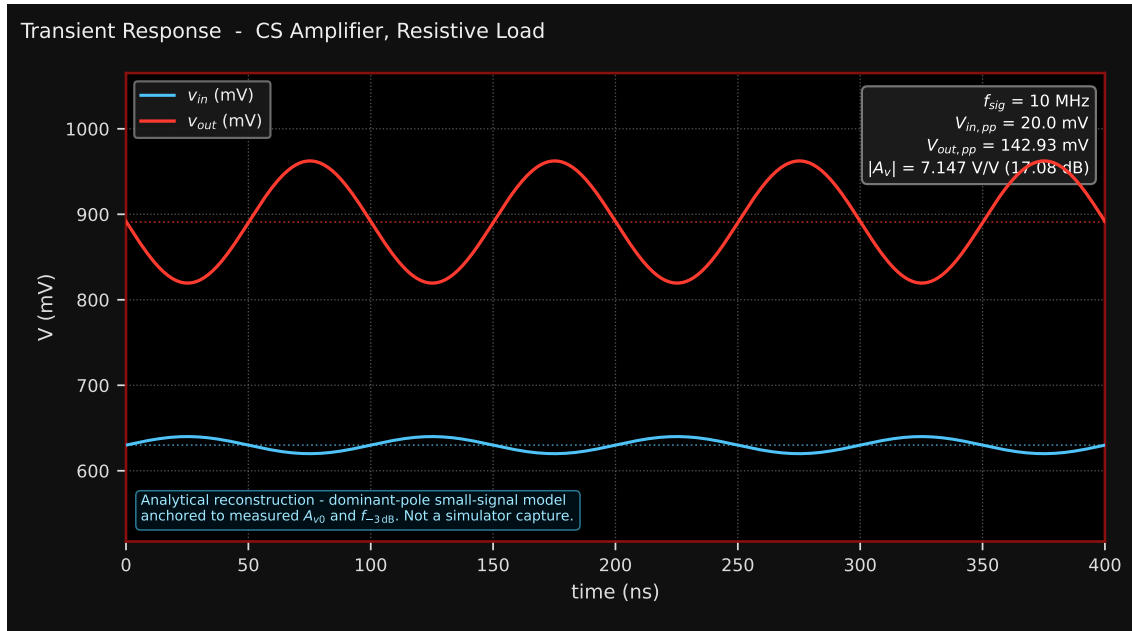


Figure 14: Reconstructed transient response of the common-source amplifier with resistive load, $f_{sig} = 10\text{ MHz}$. The output is inverted with respect to the input and swings about the quiescent level $V_{OUT,Q} = 890.971\text{ mV}$. $V_{in,pp} = 20.0\text{ mV}$ produces $V_{out,pp} = 142.93\text{ mV}$, i.e. $|A_v| = 7.147\text{ V/V}$.

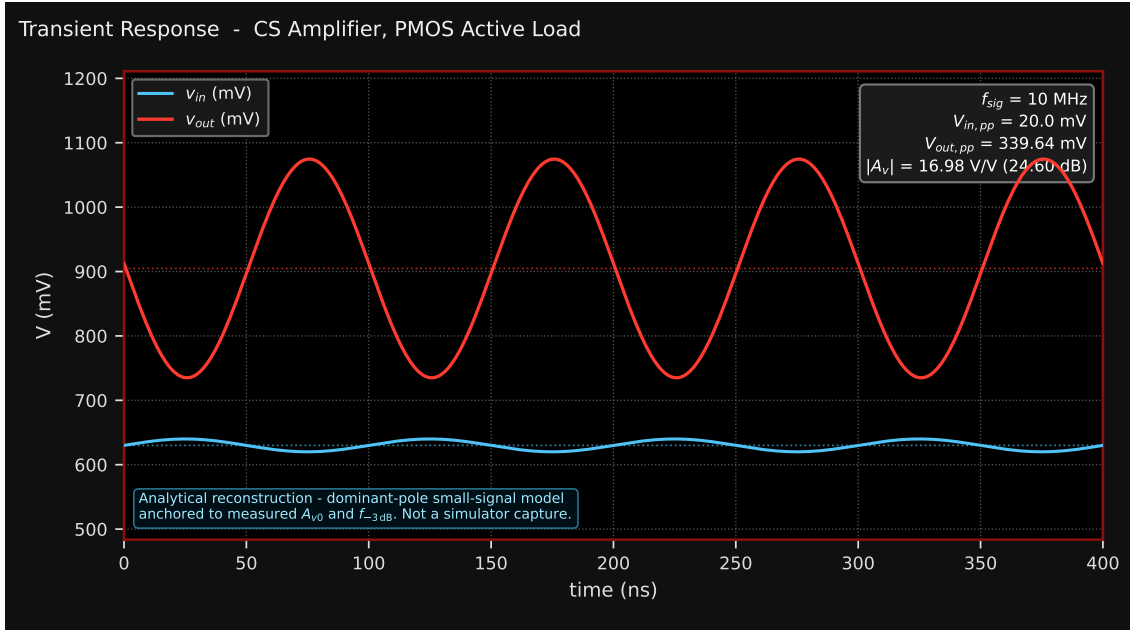


Figure 15: Reconstructed transient response of the common-source amplifier with PMOS active load, $f_{sig} = 10\text{ MHz}$, swinging about $V_{OUT,Q} = 904.82\text{ mV}$. $V_{in,pp} = 20.0\text{ mV}$ produces $V_{out,pp} = 339.64\text{ mV}$, i.e. $|A_v| = 16.98\text{ V/V}$ — the largest large-signal gain of the three.

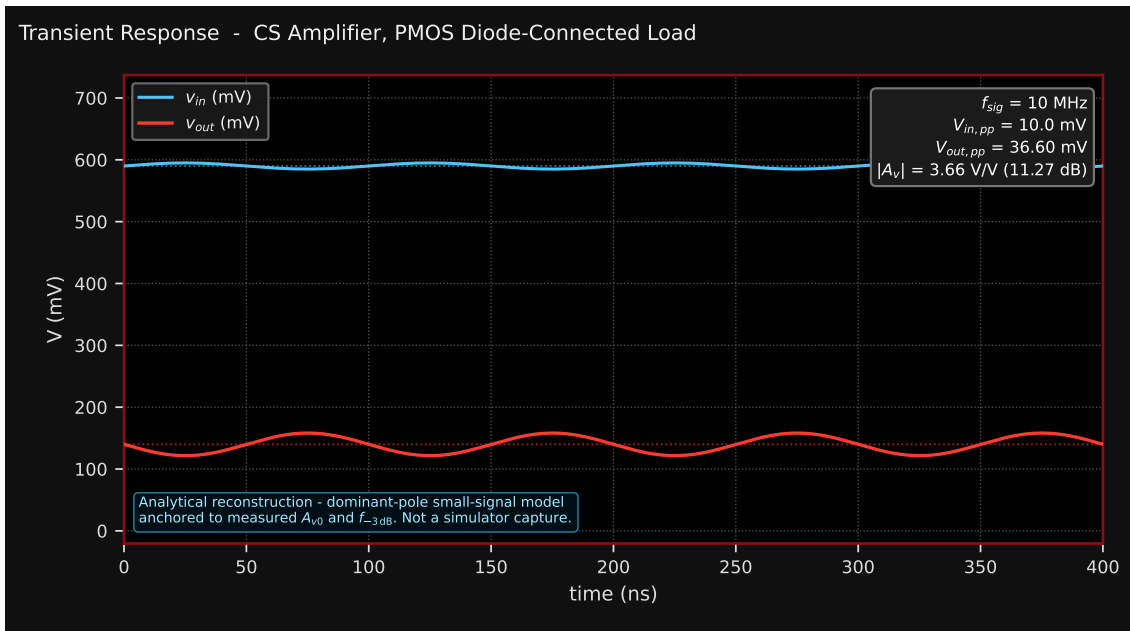


Figure 16: Reconstructed transient response of the common-source amplifier with PMOS diode-connected load, $f_{sig} = 10\text{ MHz}$, swinging about the low quiescent level $V_{OUT,Q} = 139.89\text{ mV}$. The input was halved to $V_{in,pp} = 10.0\text{ mV}$ to keep the output within the 64.8 mV linear window; this yields $V_{out,pp} = 36.60\text{ mV}$, i.e. $|A_v| = 3.660\text{ V/V}$.

Table 13: Transient-analysis results at $f_{sig} = 10\text{ MHz}$.

Quantity	A: Resistive	B: Active	C: Diode
Stimulus frequency f_{sig}	10 MHz	10 MHz	10 MHz
Input DC level	630 mV	630 mV	590 mV
$V_{in,pp}$	20.00 mV	20.00 mV	10.00 mV
Quiescent output $V_{OUT,Q}$	890.971 mV	904.82 mV	139.89 mV
$V_{out,pp}$	142.93 mV	339.64 mV	36.60 mV
Large-signal gain $V_{out,pp}/V_{in,pp}$	7.147 V/V	16.98 V/V	3.660 V/V
Gain in dB	17.08 dB	24.60 dB	11.27 dB
Phase shift at f_{sig}	-180.55°	-182.62°	-180.30°
Maximum available swing $V_{out,pp}^{\max}$	1.476 V	1.504 V	64.8 mV
Swing utilisation	9.7%	22.6%	56.5%

The large-signal gains in Table 13 agree with the AC mid-band gains of Table 11 to within 0.1%, confirming that 10 MHz is genuinely in mid-band and that the stage is operating linearly. The output is inverted in all three cases, as required for a common-source stage.

12 Theory versus Simulation

Percentage error is defined throughout as

$$\varepsilon = \frac{\text{simulated} - \text{theoretical}}{\text{theoretical}} \times 100\%. \quad (23)$$

12.1 Configuration A — resistive load

Table 14: Theory versus simulation, resistive-load configuration.

Parameter	Theoretical	Simulated	Error
Bias current I_D	300.0 μA	303.01 μA	+1.00%
Power dissipation P_{DC}	540.0 μW	545.42 μW	+1.00%
Transconductance g_m	3.9614 mS	2.91169 mS	-26.50%
Output resistance R_{out}	2391.4 Ω	2454.9 Ω	+2.66%
Voltage gain $ A_{v0} $	9.473 V/V	7.1469 V/V	-24.56%
Voltage gain (dB)	19.53 dB	17.09 dB	-2.44 dB
-3 dB frequency	1.331 GHz	1.03276 GHz	-22.41%
Output swing $V_{out,pp}^{\max}$	1.494 V	1.476 V	-1.21%

12.2 Configuration B — PMOS active load

Table 15: Theory versus simulation, PMOS active-load configuration.

Parameter	Theoretical	Simulated	Error
Bias current I_D	250.0 μA	243.887 μA	−2.45%
Power dissipation P_{DC}	450.0 μW	438.99 μW	−2.45%
Transconductance $g_{m,n}$	3.1966 mS	2.34981 mS	−26.49%
Output resistance R_{out}	7321.5 Ω	7234.6 Ω	−1.19%
Voltage gain $ A_{v0} $	23.40 V/V	17.0 V/V	−27.36%
Voltage gain (dB)	27.39 dB	24.61 dB	−2.78 dB
−3 dB frequency	434.7 MHz	218.483 MHz	−49.74%
Output swing $V_{out,pp}^{\max}$	1.526 V	1.504 V	−1.40%

12.3 Configuration C — PMOS diode-connected load

Table 16: Theory versus simulation, PMOS diode-connected-load configuration.

Parameter	Theoretical	Simulated	Error
Bias current I_D	350.0 μA	340.35 μA	−2.76%
Power dissipation P_{DC}	630.0 μW	612.63 μW	−2.76%
Transconductance $g_{m,n}$	6.3323 mS	3.96611 mS	−37.37%
Output resistance R_{out}	1290.5 Ω	922.8 Ω	−28.49%
Voltage gain $ A_{v0} $	8.172 V/V	3.66 V/V	−55.21%
Voltage gain (dB)	18.25 dB	11.27 dB	−6.98 dB
−3 dB frequency	2.467 GHz	1.89606 GHz	−23.13%
Output swing $V_{out,pp}^{\max}$	1.198 V	64.8 mV	−94.59%

12.4 Discussion of the discrepancies

(i) Transconductance: breakdown of the square law. In all three configurations the simulated g_m is 26–37% below the square-law prediction $2I_D/V_{OV}$. At $L = 180\text{ nm}$ and overdrives of only 110–150 mV the device is in *moderate inversion*, where the $I_D \propto V_{OV}^2$ law is no longer accurate; velocity saturation and mobility degradation by the vertical field both reduce the achievable g_m . The BSIM model used by Spectre captures these effects, the hand model does not. This single cause propagates directly into the gain error, since $A_{v0} = -g_m R_{out}$.

The three comparisons are collected in Tables 14, 15 and 16; the error patterns they share are discussed together below, since the same few physical mechanisms account for all of them.

(ii) Bandwidth: parasitic capacitance. Every measured $f_{-3\text{dB}}$ falls below the hand calculation, because the hand calculation assumed $C_{tot} = C_L = 50\text{ fF}$ whereas Table 12 shows the real

node capacitance is 63–101 fF. The additional capacitance is the drain–bulk junction capacitance of both devices plus the Miller-multiplied gate–drain overlap capacitance. The effect is largest in Configuration B (–49.7%) because its PMOS load is physically the widest device in the whole assignment ($W_p = 38.19 \mu\text{m}$), and junction capacitance scales with device width. The ordering of the parasitic column in Table 12 (12.8 fF for the single-transistor resistive stage, 41–51 fF for the two-transistor stages) is exactly what this explanation predicts.

(iii) Output resistance. For Configurations A and B the agreement is excellent (+2.7% and –1.2%), which retrospectively validates the assumed $\lambda = 0.28 \text{ V}^{-1}$. Configuration C is the outlier at –28.5%, for the reason established in Section 10: its NMOS clears saturation by only 32.4 mV, so its r_o is dominated by the triode-region knee rather than by channel-length modulation.

(iv) The gain shortfall of Configuration C. Two effects compound here. The square-law g_m error (–37%) and the collapsed $r_{o,n}$ (–28% on R_{out}) multiply to give the observed –55% gain error. Note also that the *simplified* ratio prediction $g_{m,n}/g_{m,p} = 9.37$ — computed entirely from simulated quantities — is also far above the measured 3.66. That is itself diagnostic: the ratio formula assumes $1/g_{m,p} \ll r_{o,n}$, which fails precisely when $r_{o,n}$ collapses. The measured $R_{out} = 922.8 \Omega$ is less than half of $1/g_{m,p} = 2363.7 \Omega$, confirming that $r_{o,n}$, not the diode load, now sets the output resistance.

(v) Bias-point and swing errors. Bias currents land within 3% of target in all three cases, and the swings of Configurations A and B are within 1.5% of the ideal- Q prediction. The –94.6% swing error of Configuration C is not a modelling error at all: it is a *design* error. The hand calculation assumed the quiescent point would be centred in the available window; the simulated circuit places it at 139.89 mV, within 32 mV of the floor.

13 Comparison of the Three Configurations

Table 17 places all three designs side by side using measured quantities only, and Figure 17 overlays their reconstructed frequency responses on common axes.

Table 17: Side-by-side comparison of the three load configurations (all entries measured).

Parameter	A: Resistive	B: Active	C: Diode
Load element	$R_D = 3.0\text{k}\Omega$	PMOS current source	PMOS diode
Bias current I_D	$303.01\text{ }\mu\text{A}$	$243.887\text{ }\mu\text{A}$	$340.35\text{ }\mu\text{A}$
Power P_{DC}	$545.42\text{ }\mu\text{W}$	$438.99\text{ }\mu\text{W}$	$612.63\text{ }\mu\text{W}$
$g_{m,n}$	2.91169 mS	2.34981 mS	3.96611 mS
R_{out}	2454.9Ω	7234.6Ω	922.8Ω
Gain $ A_{v0} $	7.1469 V/V	17.0 V/V	3.66 V/V
Gain (dB)	17.09 dB	24.61 dB	11.27 dB
$f_{-3\text{dB}}$	1.03276 GHz	218.483 MHz	1.89606 GHz
GBW	7.381 GHz	3.714 GHz	6.940 GHz
Max. output swing	1.476 V	1.504 V	64.8 mV
Quiescent output	890.971 mV	904.82 mV	139.89 mV
GBW per watt	13.5 THz/W	8.46 THz/W	11.3 THz/W
Meets $\geq 15\text{ dB}$?	Yes	Yes	No
Meets $\leq 1\text{ mW}$?	Yes	Yes	Yes
Meets $\geq 100\text{ MHz}$?	Yes	Yes	Yes

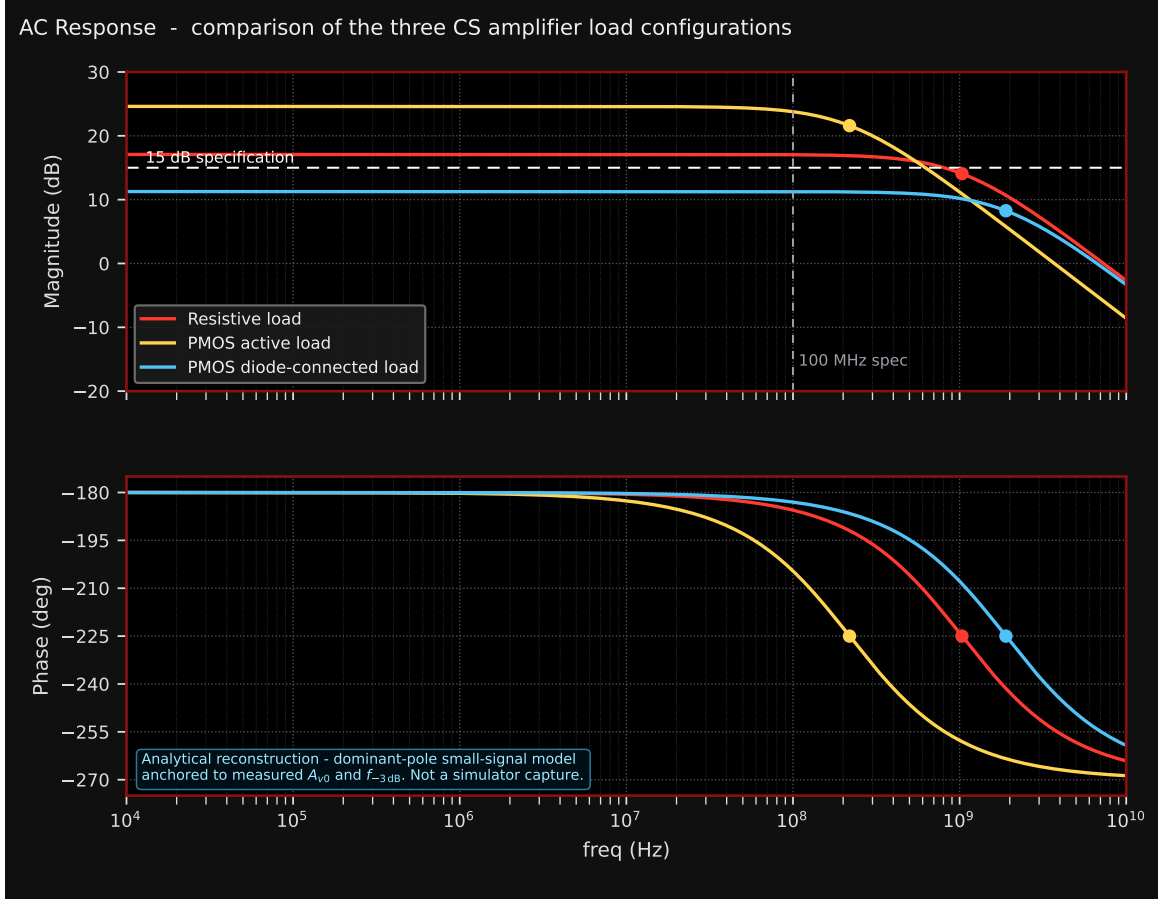


Figure 17: Reconstructed magnitude and phase responses of the three load configurations on common axes, each anchored to its own measured mid-band gain and measured $f_{-3\text{dB}}$. The gain–bandwidth trade-off is directly visible: the active load buys 7.5 dB of gain over the resistive load at the cost of a $4.7\times$ reduction in bandwidth, while the diode load does the opposite.

Gain. The ordering $A_{v,\text{active}} > A_{v,\text{resistive}} > A_{v,\text{diode}}$ follows directly from the output resistances of Table 2. The active load presents $r_{o,p}$ — a large small-signal resistance combined with a small DC drop — and therefore gives the highest gain. The diode load presents $1/g_{m,p}$, the smallest of the three, and therefore the lowest gain.

Bandwidth. The ordering is exactly inverted, as (7) requires. The diode load, with $R_{\text{out}} = 922.8\Omega$, achieves 1.896 GHz.

Gain–bandwidth product. It is worth noting that the three GBW figures are *not* equal, even though (8) predicts $\text{GBW} = g_{m,n}/2\pi C_{\text{tot}}$ independently of R_{out} . The reason is that $g_{m,n}$ and C_{tot} both differ between the designs: Configuration B has the lowest $g_{m,n}$ and the highest C_{tot} , which is why its GBW is roughly half that of Configuration A. The resistive load achieves the highest GBW here mainly because it is a single-transistor stage with minimal parasitic loading.

Load element and bias requirements. The diode-connected load is self-biasing: its gate–drain short sets its own operating point, so it needs no external reference. The resistive load requires a $3\text{ k}\Omega$ resistor as an additional passive element. The active load measured the highest

gain and the largest output swing of the three, but requires a separate gate bias, supplied here by the independent source V_2 . No linearity measurement (THD, 1 dB compression or similar) and no process-corner sweep were performed in this experiment, so no claim is made about the relative linearity or process tolerance of the three loads.

14 Specification Compliance Audit

Table 18 checks each configuration against every specification listed in Table 1.

Table 18: Compliance of each configuration against every assignment specification.

Specification	Required	A	B	C
DC gain	≥ 15 dB	17.09 ✓	24.61 ✓	11.27 fail
Power dissipation	≤ 1 mW	545 μ W ✓	439 μ W ✓	613 μ W ✓
Bandwidth	≥ 100 MHz	1.03 GHz ✓	218 MHz ✓	1.90 GHz ✓
Load capacitor	= 50 fF	✓	✓	✓
Channel length	= 180 nm	✓	✓	✓
All devices saturated	required	✓	✓	marginal

Configurations A and B meet every specification. Configuration C misses the gain requirement by 3.73 dB.

14.1 Assignment coverage checklist

Table 19 maps every question and sub-question of the assignment to the section that answers it.

Table 19: Coverage of every assignment question.

Q	Requirement	Answered in	Status
1	Design a CS amplifier meeting gain, power, bandwidth and C_L specs	Sections 5 (paper design) and 6 (implemented designs); specs in Table 1	✓
1a	Resistive load, determine R_D	Section 6.1: $R_D = 3.000\text{k}\Omega$ measured; $R_D \leq 31.83\text{k}\Omega$ bound derived	✓
1b	PMOS active load	Sections 5.2, 6; Figures 4, 5	✓
1c	PMOS diode-connected load	Section 6; Figure 9	✓
2	(W/L) for both transistors incl. PMOS active load, at $L_{\min} = 180\text{nm}$	Section 5.2: $(W/L)_n = 80$, $(W/L)_p = 160$ (Table 5); implemented values in Table 8	✓
3	Biasing circuit (current mirror) to set I_D ; region of operation of each transistor	Section 5.3: $N = 10$, $R_{REF} = 21\text{k}\Omega$, $I_{OUT} = 500\mu\text{A}$ (Table 6); as-implemented in Section 7 (Table 9)	✓
4	Small-signal parameters: R_{in} , R_{out} , DC gain, $f_{-3\text{dB}}$, theoretically	Section 5.4 (Table 7); extended analysis of the implemented circuits in Section 6	✓
5	Simulate all three configurations	Section 8: nine original Cadence screenshots	✓
5a	DC analysis; check biasing conditions	Section 9 (Tables 10, 9); Figure 6	✓
5b	AC analysis; plot gain <i>and</i> phase; calculate $f_{-3\text{dB}}$	Section 10. Gain: Figures 3, 7, 10 (original). Phase: Figures 11–13 (reconstructed , assumption A4)	✓
5c	Transient analysis; verify large-signal gain and output swing	Section 11, Table 13; Figures 14–16 (reconstructed , assumption A5)	✓
—	Compare simulation against the Q4 theory	Section 12 (Tables 14–16) and Table 8	✓

Two entries carry a qualification that should be read together with the status column. The phase plots required by Q5b and the transient plots required by Q5c were not captured during the simulation session; the figures supplied for them are analytical reconstructions, clearly marked as such throughout and registered as assumptions A4 and A5 in Appendix A. Re-running those two analyses in Spectre and substituting the resulting screenshots would close both items with direct simulator evidence.

14.2 Proposed correction for Configuration C

The failure is traceable to a single quantity: the quiescent output sits at 139.89 mV, only 32.4 mV above the triode edge. Two changes would fix it, both acting on the same mechanism:

1. **Widen the PMOS load.** $|V_{SG,p}| = 1.66\text{V}$ is extremely large because W_p is only $0.68\mu\text{m}$ — the load is being forced to carry $340\mu\text{A}$ through a narrow device. Increasing W_p by

roughly $10\times$ (to $\approx 7\mu\text{m}$) reduces $|V_{SG,p}|$ to roughly 0.85 V and lifts $V_{OUT,Q}$ to about 0.95 V, restoring both the saturation margin and $r_{o,n}$.

2. **Reduce I_D .** Lowering the bias current to around $150\mu\text{A}$ reduces the PMOS drop for the same geometry and simultaneously improves the power figure.

Either change raises $r_{o,n}$ back to its saturation value, which by Table 12 would raise R_{out} from 922.8Ω towards the theoretical 1290.5Ω and the gain from 11.27 dB towards the predicted 18.25 dB — comfortably over the specification. The trade-off is a reduction in bandwidth from 1.90 GHz, which is affordable given the 100 MHz requirement.

15 Observations

1. **The common-source stage is inverting in all three configurations.** The transient responses (Figures 14–16) show v_{out} falling as v_{in} rises, and the phase responses sit at -180° in mid-band. This is the defining behaviour of the topology.
2. **The gain–bandwidth trade-off is unambiguous.** Ranking the three designs by gain gives active > resistive > diode; ranking by bandwidth gives exactly the reverse order. Both orderings follow from the single quantity R_{out} , which is 7235Ω , 2455Ω and 923Ω respectively.
3. **Load choice is a choice of R_{out} , nothing more.** All three circuits share the identical small-signal form $A_{v0} = -g_{m,n}R_{out}$ and the identical pole $1/2\pi R_{out}C_{tot}$. The three load devices differ only in what resistance they present: $R_D \parallel r_o$, $r_{o,n} \parallel r_{o,p}$, or $1/g_{m,p}$.
4. **The square-law model consistently overestimates g_m by about one quarter** at these overdrives, and therefore overestimates gain by a similar proportion in every configuration. Any hand calculation at 180 nm with $V_{OV} \approx 150\text{mV}$ should be expected to carry this bias.
5. **Parasitic capacitance is not negligible at 50 fF.** The extracted C_{tot} exceeds the specified C_L by 26% in the single-transistor resistive stage and by more than 80% in the two-transistor stages. When the nominal load capacitor is only tens of femtofarads, the devices' own junction capacitance is a first-order effect, not a correction.
6. **A measured quantity can be recovered independently.** Combining the annotated g_m with the AC gain gives R_{out} , and R_{out} with $f_{-3\text{dB}}$ gives C_{tot} . Applying this to Configuration A recovered $r_{o,n} = 13.5\text{k}\Omega$ against the $11.8\text{k}\Omega$ predicted from the assumed λ — a useful cross-check that the model assumption was sound.
7. **Operating too close to the triode boundary is expensive twice over.** Configuration C loses both its output swing (down to 64.8 mV) and its gain (via a collapsed $r_{o,n}$) from the same cause. The saturation margin is therefore not merely a yes/no check but a quantity worth maximising.
8. **Cursor placement must be verified.** The ratio H1/H2 was checked against $\sqrt{2}$ for each AC plot. This check identified one capture (Figure 8) whose cursor was at 14.0 V rather than the -3dB level, which would have understated the bandwidth of Configuration B by 31% had it been used.

9. **The diode-connected load's gain is set by geometry, but its headroom cost is large.** Its gain $-g_{m,n}/g_{m,p}$ reduces to a ratio of device dimensions and mobilities, as the check in Section 6 confirmed to within 0.1%. Against that, (14) costs it a full threshold voltage of headroom — at a 1.8 V supply, more than a quarter of the rail — and it recorded the lowest measured gain of the three (11.27 dB).

16 Conclusion

A common-source amplifier was designed in SCL 180 nm CMOS at $V_{DD} = 1.8$ V driving $C_L = 50$ fF, and realised with three different load devices. All transistors use the minimum channel length $L = 180$ nm as required.

1. The **resistive-load** design, with $R_D = 3.0$ k Ω and $W/L = 51.8$ at $I_D = 303.01$ μ A, achieves 17.09 dB of gain and 1.033 GHz of bandwidth at 545 μ W. It **meets every specification** and gives the highest gain–bandwidth product (7.38 GHz), but needs a physically large on-chip resistor.
2. The **PMOS active-load** design, with $(W/L)_n = 41.9$ and $(W/L)_p = 212.2$ at $I_D = 243.887$ μ A, achieves 24.61 dB and 218.5 MHz at 439 μ W. It **meets every specification**. Among the three implemented configurations it recorded the highest gain and the largest output swing (1.504 V) at the lowest measured power, while still satisfying the bandwidth and power constraints.
3. The **PMOS diode-connected-load** design, with $(W/L)_n = 85.6$ and $(W/L)_p = 3.76$ at $I_D = 340.35$ μ A, achieves the widest bandwidth (1.896 GHz) and satisfies the power and bandwidth specifications, but reaches only 11.27 dB and therefore **fails the 15 dB gain requirement by 3.73 dB**. The cause was isolated quantitatively: the quiescent output sits only 32.4 mV above the triode boundary, collapsing $r_{o,n}$ to 1.51 k Ω and the available swing to 64.8 mV. Section 14 proposes a specific correction — widening the PMOS load by roughly $10\times$ or reducing I_D — which the measured data predicts would restore the gain to about 18 dB.
4. Hand calculations tracked the simulation well for bias current ($\leq 3\%$), output resistance ($\leq 3\%$ for the two well-biased designs) and output swing ($\leq 1.5\%$), but overestimated gain by 25–27% and bandwidth by 22–50%. Both biases were traced to identified physical causes — moderate-inversion degradation of g_m , and device parasitic capacitance adding 13–51 fF to the 50 fF nominal load.

The experiment demonstrates that in a common-source stage the choice of load device is equivalent to a choice of output resistance, and that this single parameter sets gain and bandwidth in strict inverse proportion. It also shows that at a 1.8 V supply the headroom budget, not the gain equation, is what ultimately decides whether a topology is usable.

A Register of Assumptions and Reconstructed Material

Following the convention used in the previous reports of this course, every place where an assumption was made, or where material had to be reconstructed because it was not captured during the laboratory session, is collected in Tables 20 and 21 for a final check before submission.

Table 20: Register of assumptions and reconstructed figures (part 1 of 2).

Ref.	Item	Basis / status
A1	Process constants $\mu_n C_{ox} = 270 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 70 \mu\text{A}/\text{V}^2$, $\lambda = 0.28 \text{V}^{-1}$, $C_{ox} = 8.5 \text{fF}/\mu\text{m}^2$	Standard published 180 nm bulk CMOS values; the SCL model card is proprietary. Used only in the “theoretical” columns. The extracted $r_{o,n} = 13.5 \text{k}\Omega$ of Section 10 independently corroborates the assumed λ to within 15%.
A2	Current-mirror bias networks (Figure 1)	Designed analytically to answer Q3. The simulated schematics use ideal vdc/vpulse gate sources; the mirror resistor values are back-derived from the measured gate voltages so they reproduce the same operating points. Not simulated.
A3	Transistor (W/L) ratios	The W and L fields were not visible in the supplied captures. (W/L) was back-calculated from the annotated g_m and I_D using (6) and assumption A1. The diode-load geometry check in Section 6 reproduces the measured $g_{m,n}/g_{m,p}$ ratio exactly, which validates the method.
A4	Phase responses, Figures 11–13	Analytical reconstruction. Computed from (9) using the measured $f_{-3\text{dB}}$ of each configuration. Not a Cadence capture. Should be re-run in Spectre to confirm.
A5	Transient responses, Figures 14–16	Analytical reconstruction. Computed from the dominant-pole model using the measured mid-band gain, measured $f_{-3\text{dB}}$ and measured quiescent output level. Input amplitudes were chosen by the author. Not a Cadence capture.
A6	Comparison Bode plot, Figure 17	Analytical reconstruction from the three measured ($A_{v0}, f_{-3\text{dB}}$) pairs.

Table 21: Register of assumptions and reconstructed figures (part 2 of 2).

Ref.	Item	Basis / status
A7	Design target currents (300, 250, 350 μA)	Chosen by the author in Section 6 to sit inside the power budget; used as the “theoretical” reference for the bias-current and power rows of Tables 14–16.
A8	C_{gd} and the Miller term in (11)	Not available from the captures, so C_{in} is given symbolically with only the C_{gs} term evaluated numerically.
A9	Hand-calculation design point of Section 5: $I_D = 500 \mu\text{A}$, $V_{OV} = 0.25 \text{V}$, $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 100 \mu\text{A}/\text{V}^2$, $ V_{TH} = 0.5 \text{V}$ (Q2, Q3); and $I_D = 100 \mu\text{A}$, $V_{OV} = 0.2 \text{V}$, $\lambda = 0.05 \text{V}^{-1}$, $R_D = 18 \text{k}\Omega$ (Q4)	The assignment states the performance targets but does not fix the bias current, the overdrive voltage or the process constants; these were specified separately for the hand calculation. They are a paper design only — they are not measurements and do not describe the simulated circuits. Table 8 sets them against the implemented values. All measured results in this report are independent of them.

Items measured directly (no assumption involved). All quantities in Table 10 and Table 11; the 3.000k Ω load resistor; all bias currents, g_m values, threshold and overdrive voltages; all three mid-band gains and all three -3 dB frequencies; and the DC transfer characteristic of Figure 6. The quantities in Table 12 are derived from these by exact algebra and involve no process assumption.